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1 Main Figures

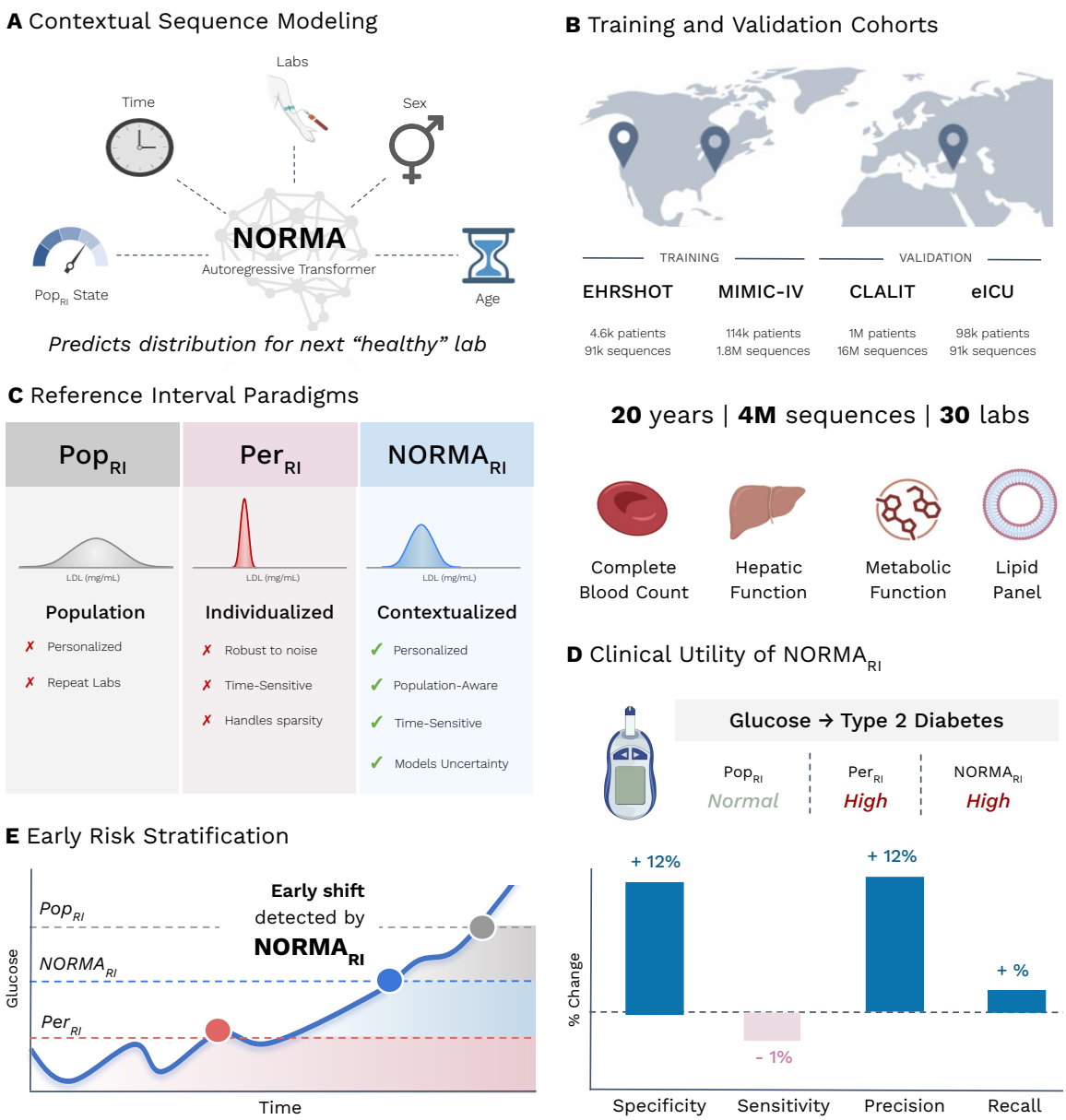

Figure 1: Overview of NORMA. (A) Contextual sequence modeling: NORMA is an autoregressive transformer that integrates labs, time, sex, age, and population-level state to predict the distribution of the next healthy lab value. (B) Training and validation cohorts spanning 20 years, 4M sequences, and 30 analytes across four datasets. (C) Reference interval paradigms: population (PopRI), individualized (PerRI), and contextualized (NORMA_{RI}). (D) Clinical utility of NORMA_{RI} for outcome prediction. (E) Early risk stratification via detection of shifts before

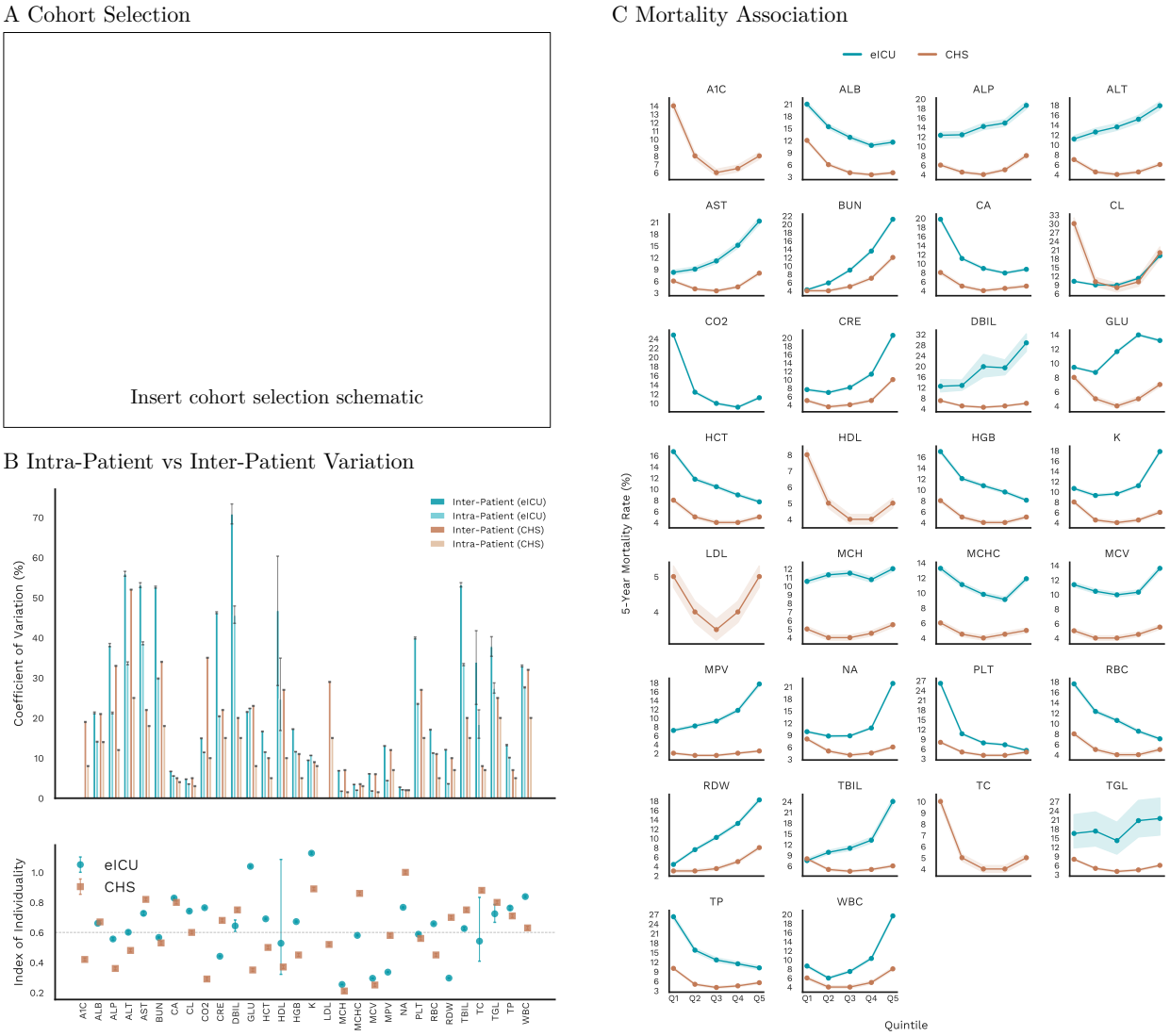

Figure 2: Detection of within-person abnormalities by personalized intervals. (A) Cohort selection schematic. (B) Intra-patient vs inter-patient coefficient of variation and index of individuality across 30 analytes. (C) Five-year mortality rate by quintile of raw analyte value for eICU and CHS cohorts.

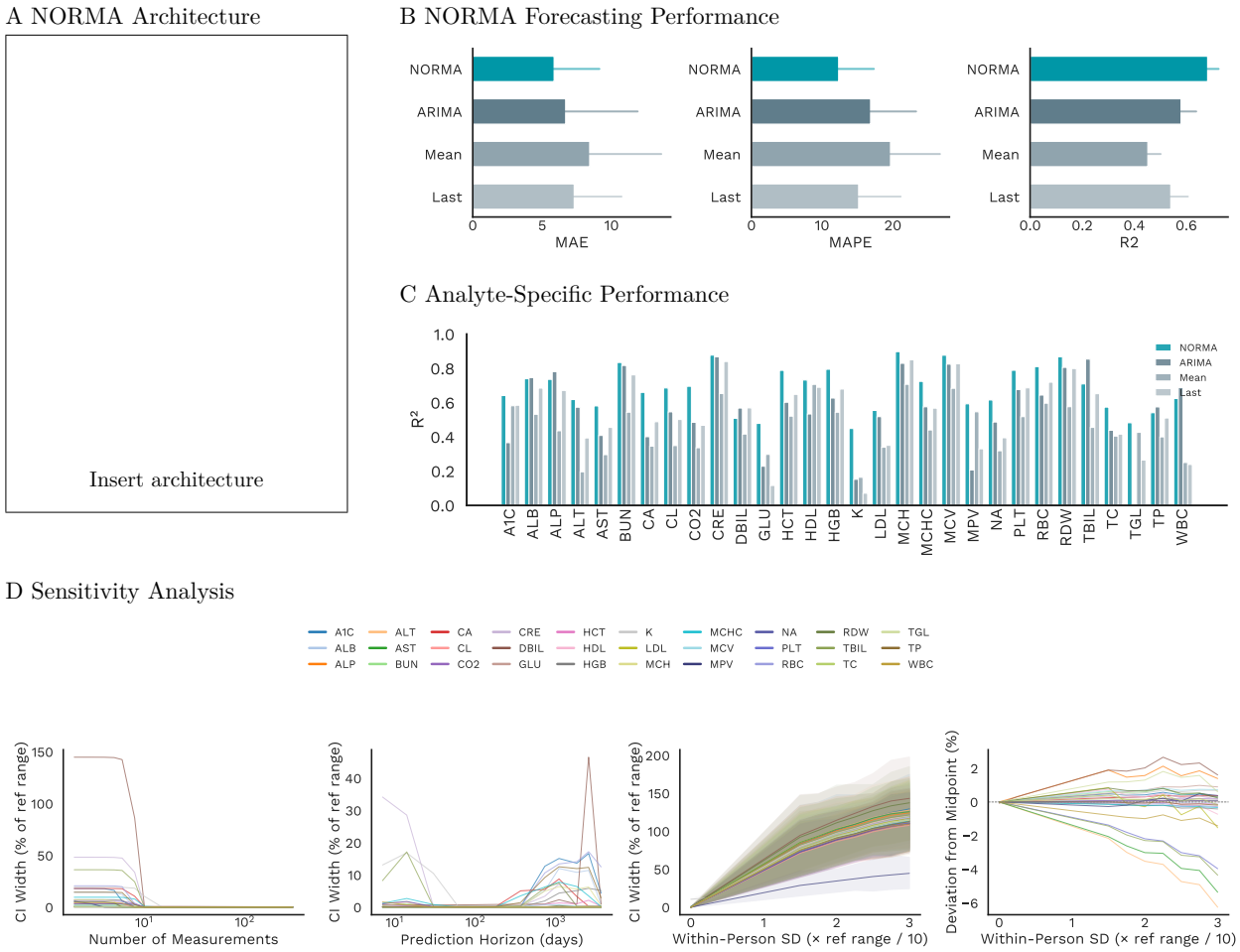

Figure 3: NORMA architecture and predictive performance. (A) Model architecture: decoder-only transformer with context, history, and query tokens producing quantile predictions. (B) Forecasting accuracy (MAE, MAPE, R^2) compared with ARIMA, patient mean, and last-value baselines. (C) Per-analyte R^2 across 30 laboratory tests. (D) Sensitivity analysis: effect of history length, prediction horizon, and within-person variability on prediction interval width.

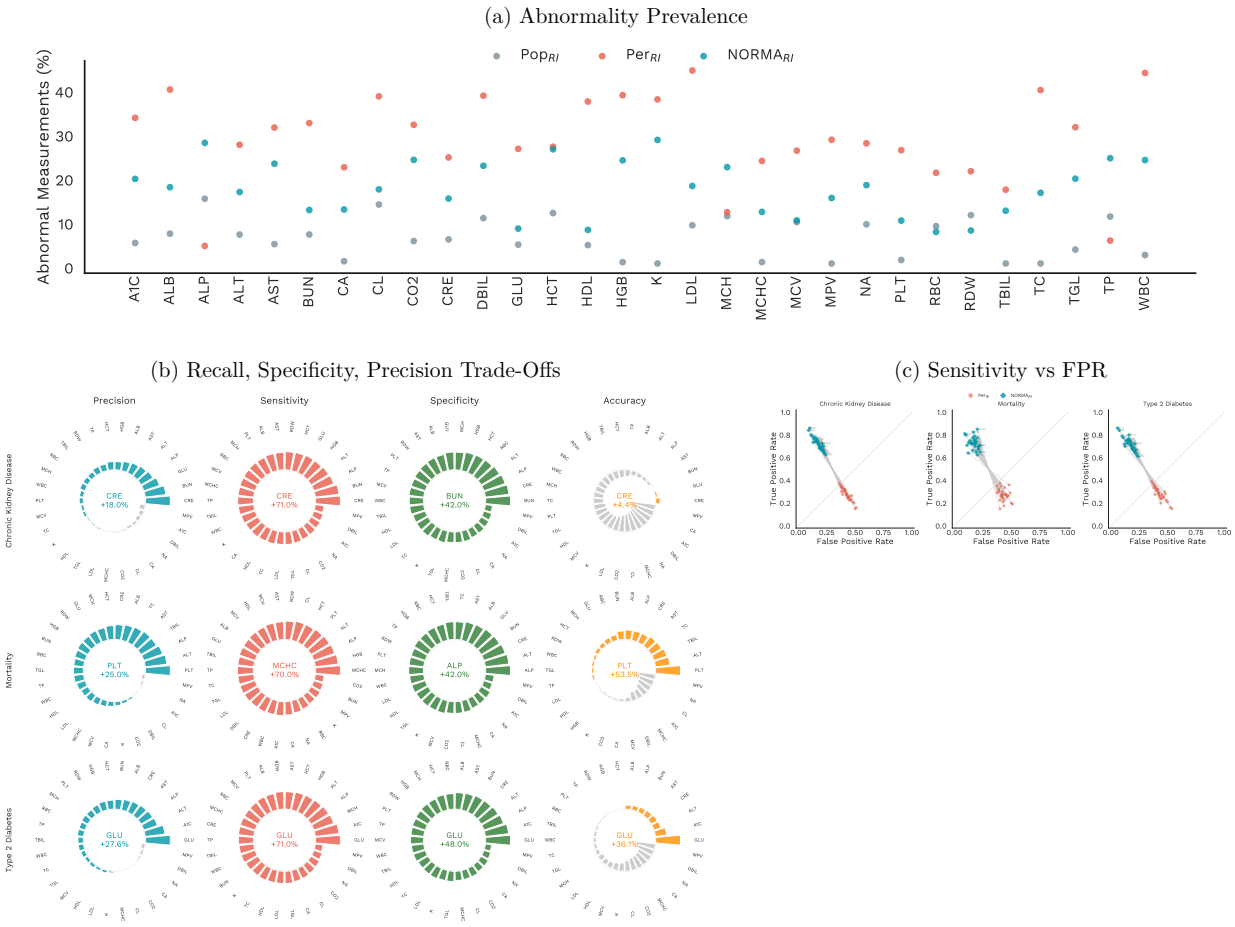

Figure 1: **Clinical outcome prediction in CHS.** (A) Abnormality prevalence by analyte across reference interval paradigms. (B–D) Performance metrics for NORMA$_{RI}$ versus Per$_{RI}$, restricted to Pop$_{RI}$-normal measurements.

Figure 4: Time-dependent discrimination for clinical outcomes in Clalit Health Services. (A) Abnormality prevalence across reference interval frameworks (PopRI, PerRI, NORMA$_{\mathrm{RI}}$). (B) Albumin reclassification across paradigms. (C) Recall, specificity, and precision trade-offs (circos plots) for mortality, CKD, and type 2 diabetes. (D) Time-to-event discrimination (AUC) by analyte and method.

Figure 5: Clinical outcome prediction in eICU. (A) Reclassification rate among PopRI-normal measurements by PerRI and NORMA_{RI}. (B) Cox proportional hazard ratios for mortality (significant associations only). (C) Sensitivity vs false positive rate for acute kidney injury, mortality, prolonged ICU stay, and sepsis. (D) Lead time: how much earlier NORMA flags abnormalities compared with PopRI. (E) Recall, specificity, precision, and accuracy trade-offs by outcome.

2 Supplemental Figures

2.1 NORMA Gaussian Model Performance

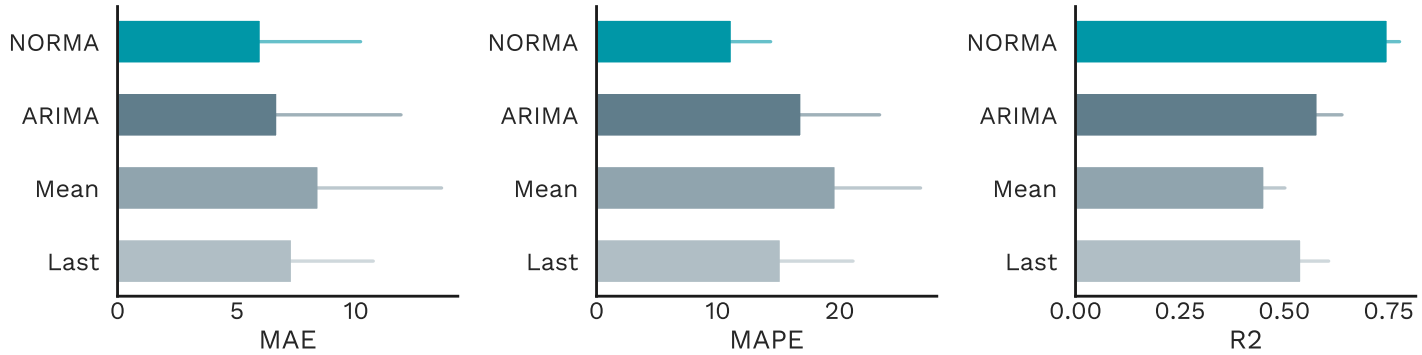

Figure 6: Supplementary Figure 1. Forecasting accuracy of NORMA (Gaussian loss) compared with baselines.

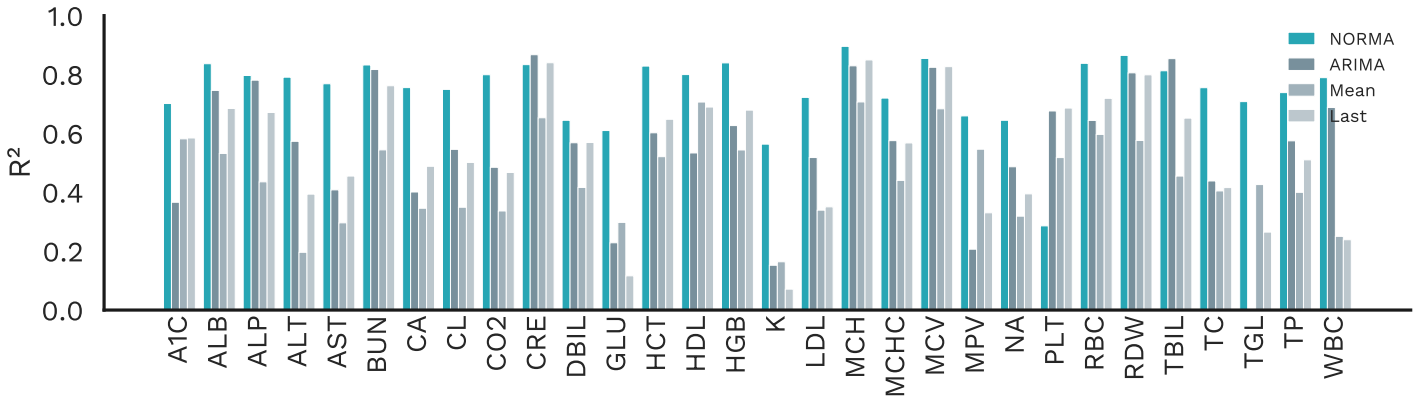

Figure 7: Supplementary Figure 2. Per-analyte R^2 for NORMA (Gaussian) and baselines.

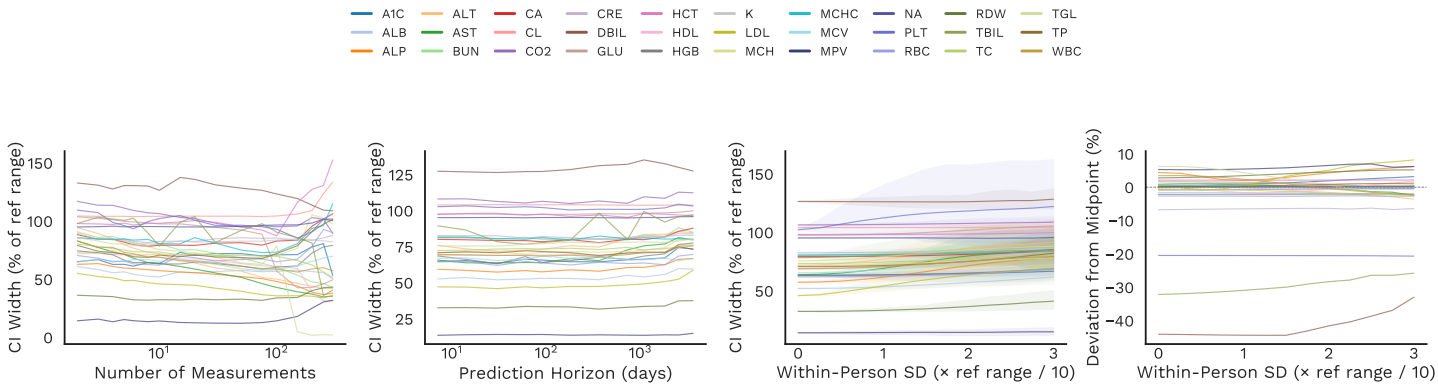

Figure 8: Supplementary Figure 3. Sensitivity analysis for NORMA (Gaussian).

2.2 Validation — eICU

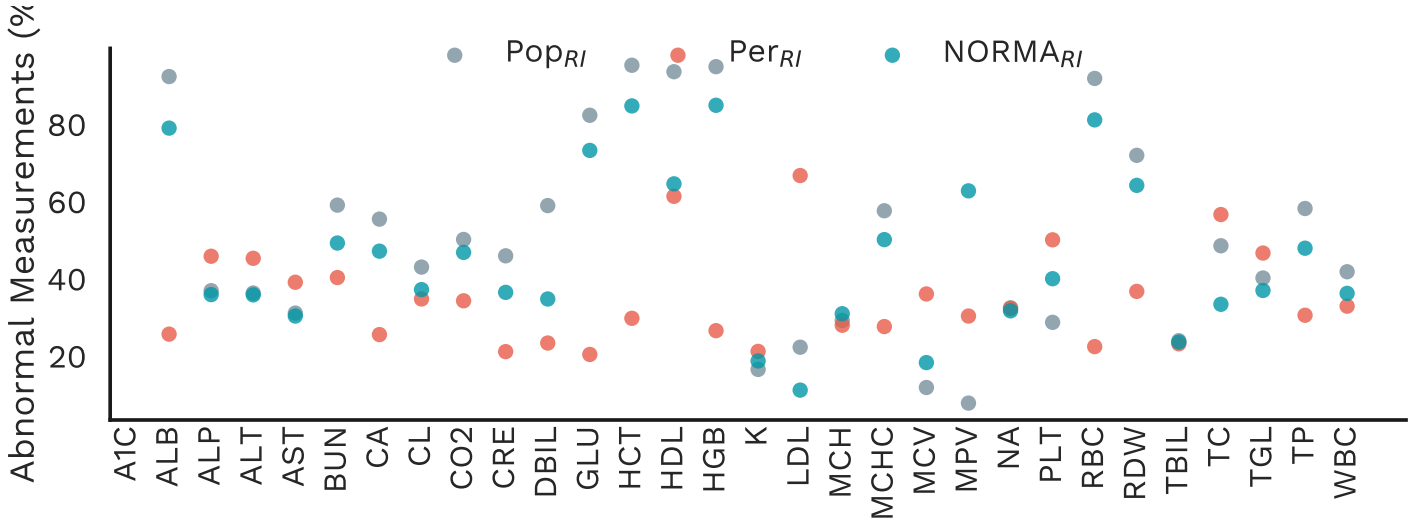

Figure 9: Supplementary Figure 4. Abnormality prevalence across reference interval frameworks in eICU.

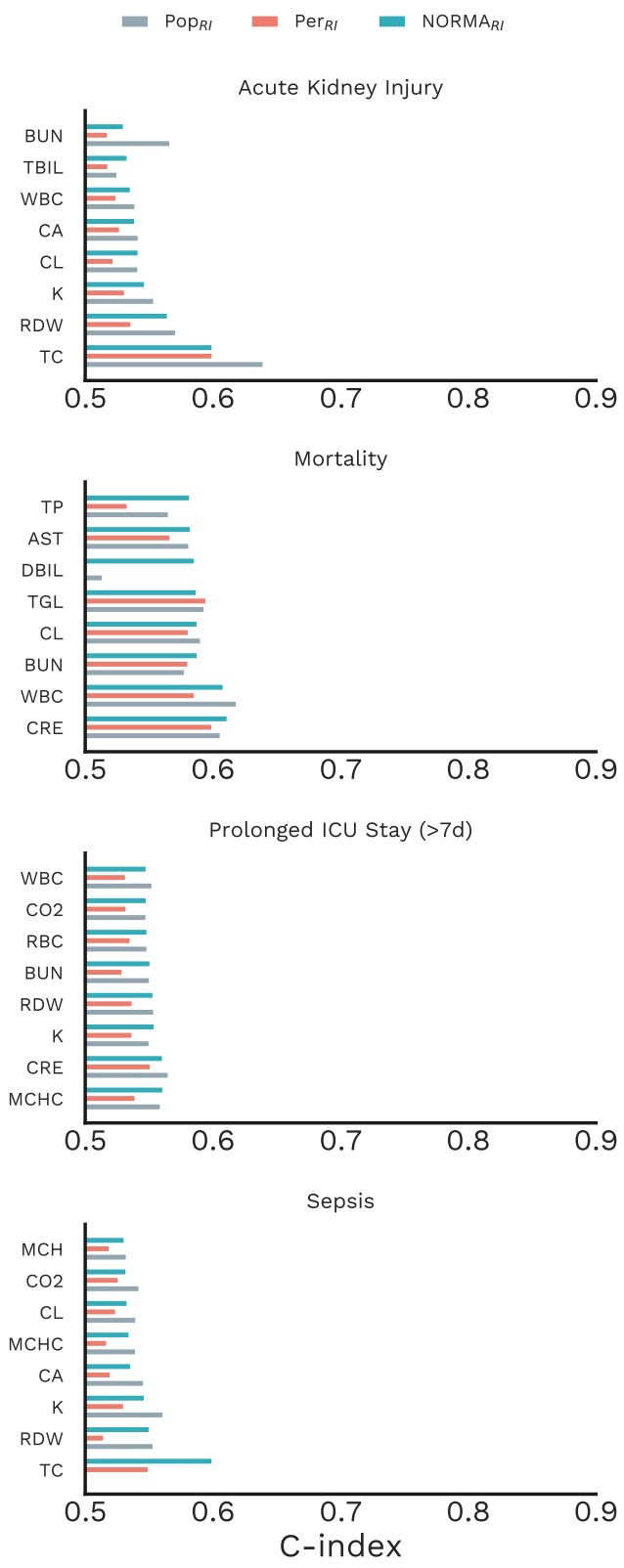

Figure 10: Supplementary Figure 5. Concordance index for Cox time-to-event models by analyte and method in eICU.

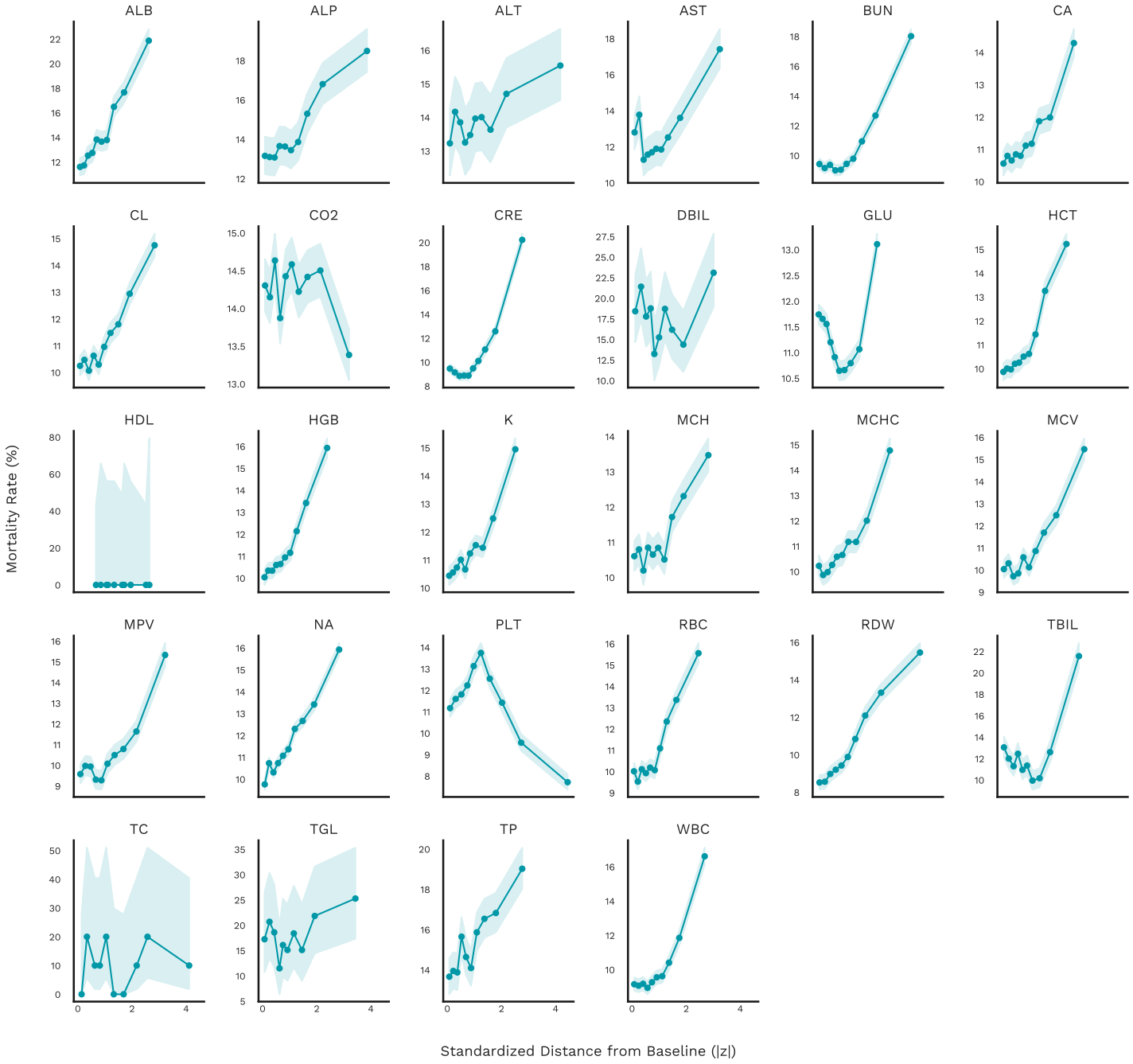

Figure 11: Supplementary Figure 6. Mortality rate by standardized deviation from personal baseline in eICU.

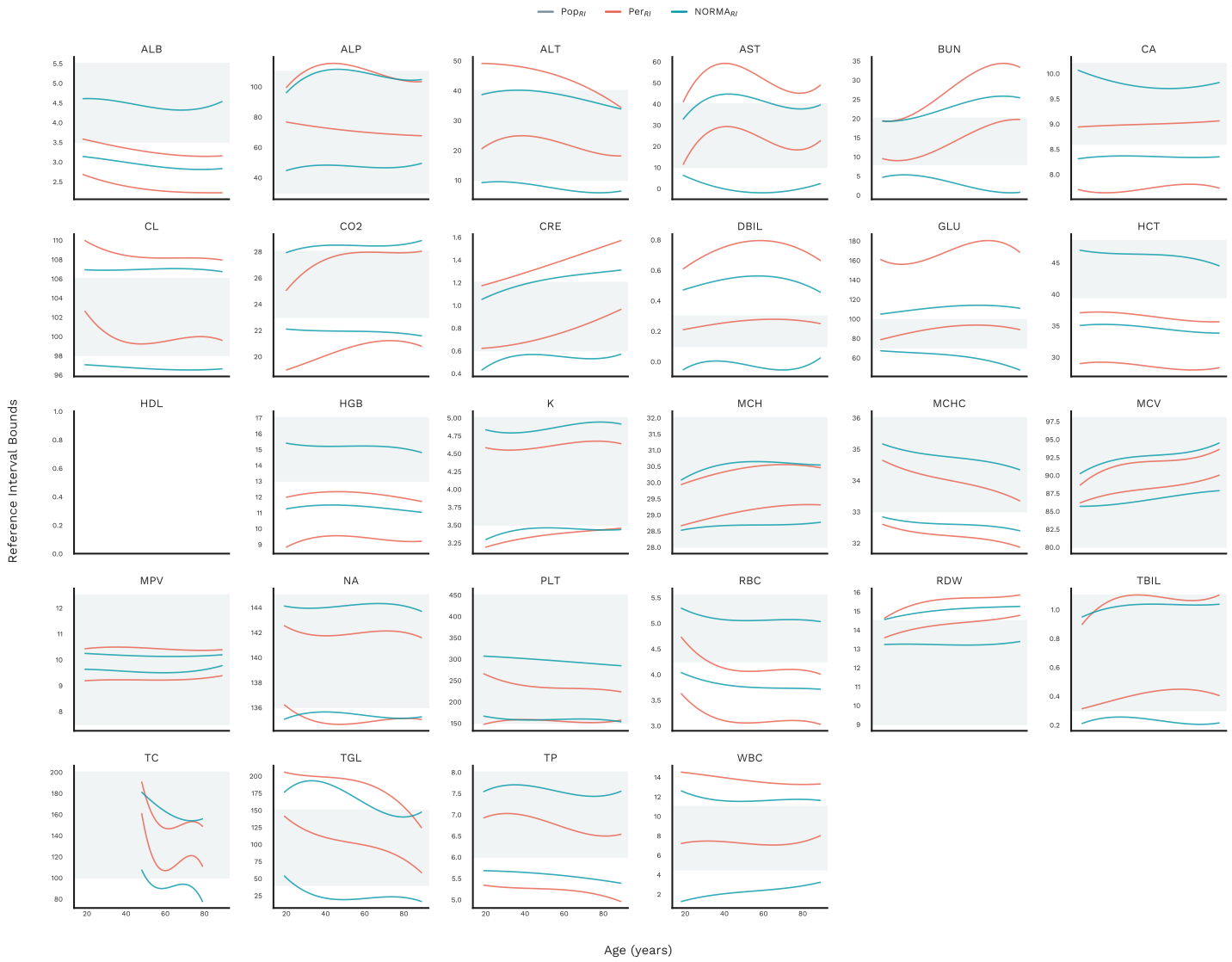

Figure 12: Supplementary Figure 7. Age-dependent reference interval trends in eICU.

2.3 Validation — CHS

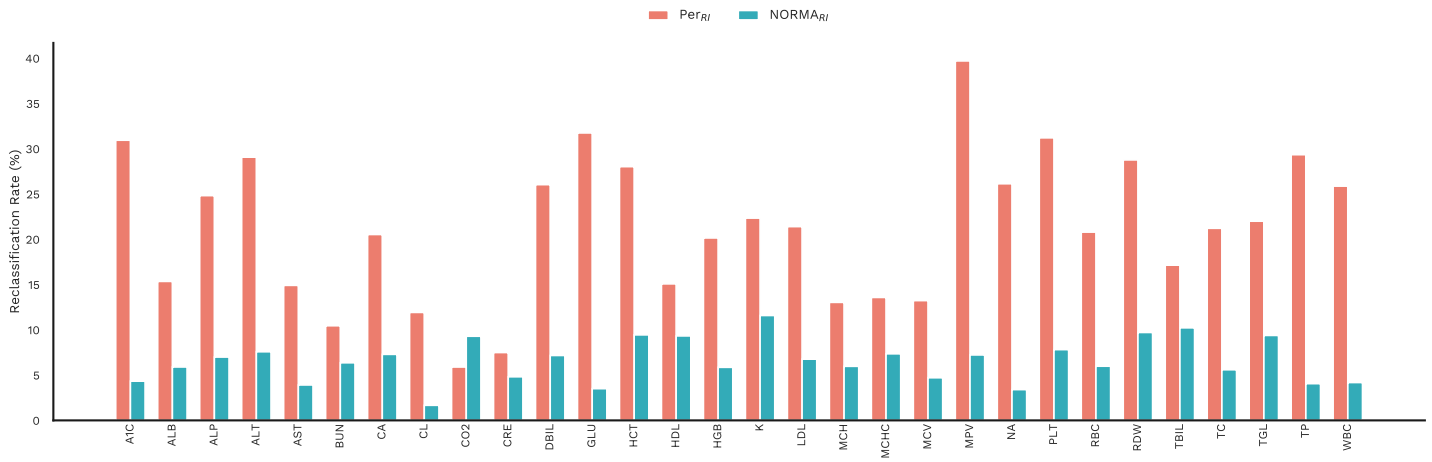

Figure 13: Supplementary Figure 8. Reclassification rate among PopRI-normal measurements in CHS.

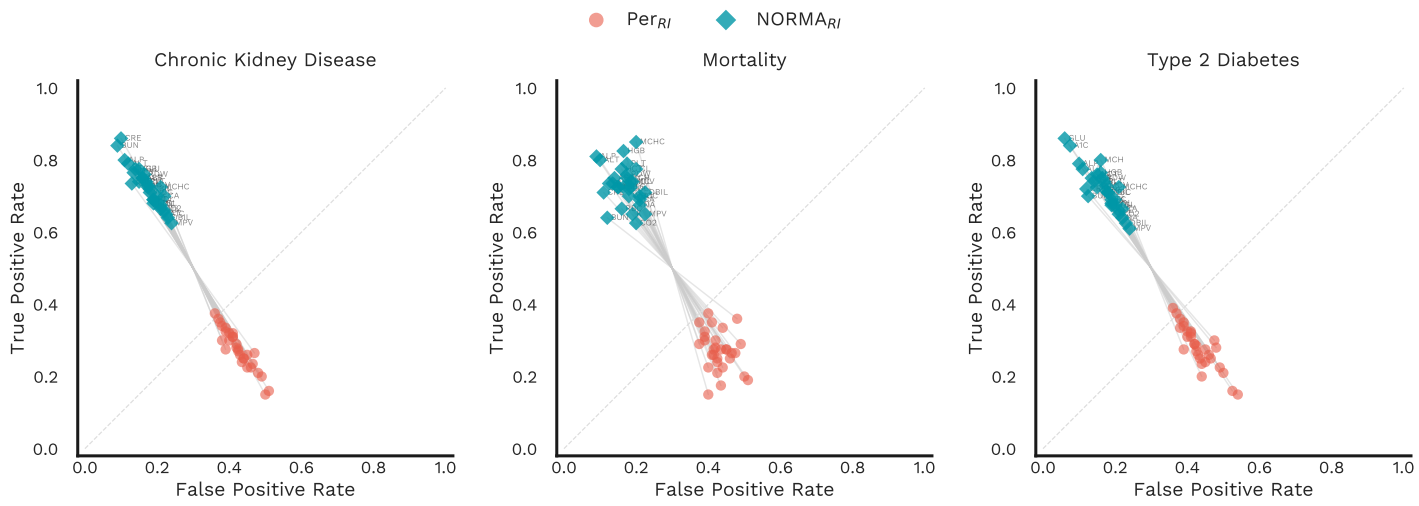

Figure 14: Supplementary Figure 9. Sensitivity vs false positive rate by outcome in CHS.

3 Supplemental Tables

3.1 Cohort Description

Analyte	Full Name	Unit	Population RI
A1C	Hemoglobin A1c	%	4.0–5.6
ALB	Albumin	g/dL	3.5–5.5
ALP	Alkaline Phosphatase	U/L	30–110
ALT	Alanine Aminotransferase	U/L	10.0–40.0
AST	Aspartate Aminotransferase	U/L	10.0–40.0
BUN	Blood Urea Nitrogen	mg/dL	8–20
CA	Calcium	mg/dL	8.6–10.2
CL	Chloride	mEq/L	98–106
CO2	Bicarbonate	mEq/L	23–28
CRE	Creatinine	mg/dL	F: 0.5–1.1, M: 0.7–1.3
DBIL	Direct Bilirubin	mg/dL	0.1–0.3
GLU	Glucose	mg/dL	70–99
HCT	Hematocrit	%	F: 37–47, M: 42–50
HDL	HDL Cholesterol	mg/dL	40–100
HGB	Hemoglobin	g/dL	F: 12.0–16.0, M: 14–18
K	Potassium	mEq/L	3.5–5.0
LDL	LDL Cholesterol	mg/dL	0–130
MCH	Mean Corpuscular Hemoglobin	pg	28–32
MCHC	MCH Concentration	g/dL	33–36
MCV	Mean Corpuscular Volume	fL	80–98
MPV	Mean Platelet Volume	fL	7.5–12.5
NA	Sodium	mEq/L	136–145
PLT	Platelet Count	$10^3/\mu$L	150–450
RBC	Red Blood Cell Count	$10^6/\mu$L	F: 4.0–5.2, M: 4.5–5.9
RDW	Red Cell Distribution Width	%	9–14.5
TBIL	Total Bilirubin	mg/dL	0.3–1.1
TC	Total Cholesterol	mg/dL	100–200
TGL	Triglycerides	mg/dL	40–150
TP	Total Protein	g/dL	6.0–8.0
WBC	White Blood Cell Count	$10^3/\mu$L	4.5–11.0

Table 1: Analyte Reference

	EHRSHOT			MIMIC-IV			eICU			CHS	
Patients	5,676			179,601			98,432			903,0	
Sequences	122,350			3,285,245			1,135,814			16,301,	
Age	54.9 ± 17.2			58.2 ± 19.0			63.3 ± 16.1			—	
Sex	52% F / 48% M			54% F / 46% M			46% F / 54% M			—	
Time	2.5 [0.3, 7.2] yr			1.0 [0.1, 4.1] yr			6 [4, 10] days			—	
Analyte	N	Value	Freq	N	Value	Freq	N	Value	Freq	N	
A1C	1,745	6.3 ± 1.2	7 ± 11	39,996	6.6 ± 1.3	5 ± 5	—	—	—	336,494	7.0
ALB	4,282	3.3 ± 0.8	28 ± 46	55,220	3.8 ± 0.8	8 ± 13	19,075	2.6 ± 0.7	14 ± 11	790,490	4.3
ALP	4,206	112.5 ± 58.8	26 ± 43	78,930	107.0 ± 58.4	8 ± 14	13,582	97.0 ± 43.2	13 ± 9	1,099,044	75.1 ±
ALT	4,219	37.8 ± 25.2	26 ± 43	92,599	30.2 ± 21.9	9 ± 15	12,425	33.8 ± 23.4	13 ± 9	1,233,902	20.9 ±
AST	4,220	31.9 ± 19.5	26 ± 43	90,639	32.7 ± 21.4	9 ± 15	12,704	37.1 ± 25.6	13 ± 9	1,236,781	21.7
BUN	5,180	23.0 ± 15.2	34 ± 55	153,329	22.2 ± 14.3	13 ± 21	57,540	25.3 ± 15.5	16 ± 13	1,251,453	15.2
CA	5,162	8.8 ± 0.7	34 ± 57	121,478	8.8 ± 0.7	12 ± 20	57,067	8.4 ± 0.7	16 ± 13	909,036	9.4
CL	5,172	102.4 ± 5.0	34 ± 57	148,304	101.8 ± 5.2	13 ± 22	59,172	103.8 ± 6.2	16 ± 14	31,171	102.7
CO2	5,187	25.9 ± 3.8	34 ± 57	145,386	25.6 ± 4.1	13 ± 21	70,208	25.0 ± 4.7	21 ± 21	299	32.4 ±
CRE	5,161	1.1 ± 0.5	34 ± 56	156,044	1.1 ± 0.6	13 ± 20	54,820	1.1 ± 0.6	16 ± 13	1,310,465	0.8
DBIL	1,429	0.3 ± 0.3	10 ± 13	6,635	0.7 ± 0.9	4 ± 4	1,147	0.6 ± 0.5	12 ± 7	—	
GLU	5,150	124.9 ± 39.9	34 ± 58	147,195	118.3 ± 40.4	11 ± 19	90,958	139.6 ± 44.3	45 ± 59	1,351,168	101.2 ±
HCT	5,470	32.2 ± 6.8	35 ± 61	164,903	33.6 ± 6.8	12 ± 20	60,110	30.1 ± 5.9	17 ± 14	1,407,912	40.6
HDL	1,549	53.6 ± 17.5	7 ± 9	49,011	55.8 ± 17.7	5 ± 4	11	26.0 ± 14.3	10 ± 3	1,002,204	50.4 ±
HGB	5,369	10.7 ± 2.3	33 ± 58	162,168	11.0 ± 2.3	12 ± 19	60,482	9.9 ± 2.0	17 ± 14	1,408,118	13.4
K	5,236	4.1 ± 0.5	38 ± 67	150,333	4.2 ± 0.5	13 ± 22	66,569	4.0 ± 0.6	18 ± 16	1,182,544	4.5
LDL	821	94.9 ± 34.9	7 ± 9	48,688	104.2 ± 37.8	5 ± 4	3	92.1 ± 56.2	11 ± 4	1,187,844	111.0 ±
MCH	5,294	30.4 ± 2.7	33 ± 57	161,671	29.7 ± 2.8	12 ± 19	48,432	29.7 ± 2.1	15 ± 12	1,250,759	28.9
MCHC	5,317	33.2 ± 1.4	33 ± 57	161,806	32.6 ± 1.6	12 ± 19	51,710	32.8 ± 1.3	15 ± 12	1,407,568	33.1
MCV	5,304	91.5 ± 6.9	33 ± 57	161,661	91.2 ± 7.3	12 ± 19	50,841	90.4 ± 5.7	15 ± 12	1,400,648	87.1
MPV	371	9.6 ± 1.6	12 ± 18	—	—	—	37,083	9.7 ± 1.4	15 ± 11	—	
NA	5,193	137.4 ± 4.2	35 ± 59	149,303	138.7 ± 4.2	13 ± 22	62,419	138.6 ± 5.0	17 ± 15	1,187,537	140.3
PLT	5,320	198.0 ± 111.4	33 ± 58	162,308	229.4 ± 113.3	12 ± 20	53,371	199.0 ± 98.4	15 ± 12	1,407,716	251.1 ±
RBC	5,312	3.5 ± 0.8	33 ± 58	161,825	3.7 ± 0.8	12 ± 19	53,584	3.4 ± 0.7	15 ± 12	1,404,794	4.7
RDW	5,315	16.1 ± 2.9	33 ± 56	161,640	15.1 ± 2.4	12 ± 19	48,715	15.8 ± 2.1	15 ± 11	1,403,339	13.6
TBIL	4,169	0.6 ± 0.4	26 ± 43	73,002	0.7 ± 0.6	8 ± 15	12,738	0.7 ± 0.5	13 ± 9	815,774	0.6
TC	1,635	173.5 ± 45.0	8 ± 12	49,762	187.5 ± 44.6	5 ± 4	35	108.3 ± 44.0	10 ± 5	1,260,730	188.9 ±
TGL	4	165.7 ± 100.8	2 ± 1	48,445	140.4 ± 77.5	5 ± 4	312	144.9 ± 68.0	10 ± 4	1,244,518	134.0 ±
TP	4,213	6.6 ± 1.0	26 ± 43	18,673	7.0 ± 0.7	4 ± 4	15,134	5.8 ± 1.0	14 ± 10	791,481	7.3
WBC	5,345	7.8 ± 4.6	33 ± 57	164,291	7.7 ± 4.0	12 ± 20	52,141	10.6 ± 4.6	15 ± 12	1,409,027	7.1

Table 2: Cohort Summary

Component	Design Choice	Description
Time embedding	Time2Vec	Encodes days since first measurement using learned linear and sinusoidal functions. Captures long-term temporal trends and periodic patterns.
	Delta-t	Encodes log-compressed time difference between consecutive measurements. Represents irregular sampling.
Health state encoding	Binary	Encodes each value as within or outside the population reference range. Does not distinguish high vs. low.
	Ternary	Encodes whether the value is low, normal, or high relative to the reference range. Handles asymmetric risk patterns.
Age encoding	Raw values	Uses age directly as numeric input. Assumes linear effects.
	Binned	Discretizes age into groups (e.g., decades) and maps each bin to a learned embedding. Captures nonlinear and threshold effects.
Lab sequence encoding	Raw values	Uses observed biomarker values directly. Preserves absolute clinical scale and keeps preprocessing minimal.
	Within-sequence normalization	Per-sequence z-scoring with learnable scale and shift, then denormalized at output. Stabilizes optimization and improves cross-analyte sharing.
Training objective	Gaussian NLL	Predicts mean and log-variance under a normal distribution. Aligns with conventional RI assumptions. May underperform for skewed or heavy-tailed labs.
	Quantile (pinball) loss	Predicts fixed quantiles (2.5, 25, 50, 75, 97.5%). Distribution-free, robust to skewness and heteroscedasticity.

3.2 Prediction Performance

	NORMA Gaussian			NORMA Quantile				
Analyte	MAE	MAPE	R2	MAE	MAPE	R2	MAE	
A1C	0.4	5.7	0.70	0.4	6.1	0.64	0.5	
ALB	0.3	7.7	0.84	0.3	8.5	0.74	0.3	
ALP	15.6	15.2	0.80	16.0	15.4	0.74	24.3	
ALT	6.2	24.7	0.79	7.2	27.8	0.62	14.5	
AST	6.0	21.0	0.77	7.0	23.5	0.58	20.4	
BUN	3.3	17.2	0.83	3.3	18.4	0.83	4.4	
CA	0.3	3.0	0.76	0.3	3.3	0.66	0.4	
CL	1.8	1.8	0.75	1.9	1.8	0.69	2.4	
CO2	1.3	5.4	0.80	1.5	6.3	0.70	2.1	
CRE	0.1	12.3	0.83	0.1	10.6	0.88	0.2	
DBIL	0.3	36.5	0.64	0.3	48.0	0.51	0.4	
GLU	14.0	11.3	0.61	15.7	12.9	0.48	21.1	
HCT	2.0	6.1	0.83	2.1	6.2	0.79	3.0	
HDL	6.0	11.1	0.80	6.5	12.1	0.73	8.8	
HGB	0.7	6.2	0.84	0.7	6.3	0.79	1.0	
K	0.3	6.3	0.56	0.3	6.7	0.45	0.4	
LDL	15.0	20.0	0.72	17.7	21.8	0.55	18.6	
MCH	0.6	2.1	0.90	0.5	1.8	0.90	0.6	
MCHC	0.6	1.9	0.72	0.6	1.8	0.72	0.6	
MCV	1.9	2.2	0.85	1.6	1.8	0.88	1.7	
MPV	0.7	6.7	0.66	0.6	6.5	0.59	0.9	
NA	1.8	1.3	0.64	1.8	1.3	0.62	2.2	
PLT	52.8	19.3	0.29	30.3	13.5	0.79	36.6	
RBC	0.2	6.4	0.84	0.2	6.3	0.81	0.3	
RDW	0.5	3.4	0.86	0.5	3.0	0.87	0.7	
TBIL	0.1	22.5	0.81	0.2	28.0	0.71	0.3	
TC	17.0	9.8	0.76	20.3	12.0	0.57	24.8	
TGL	28.7	22.8	0.71	36.2	28.7	0.48	—	
TP	0.3	4.7	0.74	0.4	5.7	0.54	0.5	
WBC	1.3	19.6	0.79	1.5	24.2	0.62	2.1	
Overall (Train)	6.0 [3.2, 11.1]	10.9 [8.8, 14.0]	0.76 [0.72, 0.80]	5.7 [2.9, 10.1]	11.9 [8.6, 14.7]	0.70 [0.66, 0.75]	6.8 [2.5, 10.2]	15
Overall (Val)	6.0 [2.6, 10.7]	11.1 [7.9, 13.2]	0.75 [0.69, 0.78]	5.8 [2.9, 8.7]	12.6 [8.7, 15.6]	0.69 [0.66, 0.73]	6.8 [3.6, 10.0]	15
Overall (Test)	6.0 [2.6, 10.2]	11.1 [8.5, 14.3]	0.75 [0.70, 0.78]	5.9 [3.6, 9.1]	12.3 [7.9, 17.3]	0.68 [0.64, 0.72]	6.7 [4.3, 11.9]	16

Table 3: Analyte Performance

	History Length		Prediction Horizon		Within-Person Variability	
Analyte	CI (% ref)	q50 (%)	CI (% ref)	q50 (%)	CI (% ref)	q50 (%
A1C	0.1 to 18.1	-0.0 to 0.0	0.1 to 16.7	-0.0 to 0.0	0.1 to 129.4	-0.0 to 0.6
ALB	0.1 to 14.5	-0.0 to 0.0	0.1 to 12.0	-0.0 to 0.0	0.1 to 118.6	-0.1 to 0.2
ALP	0.0 to 0.4	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 124.2	-0.0 to 2.1
ALT	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 127.7	-6.3 to -0.0
AST	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 125.8	-5.4 to -0.0
BUN	0.0 to 2.4	-0.0 to 0.0	0.0 to 1.2	-0.0 to 0.0	0.0 to 120.0	-0.0 to 0.8
CA	0.1 to 18.1	-0.0 to 0.0	0.1 to 8.8	-0.0 to 0.0	0.1 to 109.9	-0.0 to 0.4
CL	0.0 to 3.6	-0.0 to 0.0	0.0 to 1.6	-0.0 to 0.0	0.0 to 107.8	-0.0 to 0.1
CO2	0.0 to 5.8	-0.0 to 0.0	0.1 to 0.4	-0.0 to -0.0	0.1 to 110.1	-0.0 to 0.2
CRE	0.2 to 48.3	-0.0 to 0.1	0.3 to 34.2	-0.0 to 0.1	0.3 to 118.3	-0.0 to 0.5
DBIL	0.4 to 144.8	-0.1 to 0.7	0.7 to 46.5	-0.2 to -0.1	0.7 to 143.3	-0.1 to 2.7
GLU	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.8	-0.0 to 0.0	0.0 to 113.5	-0.0 to 1.0
HCT	0.0 to 3.6	-0.0 to 0.0	0.0 to 2.2	-0.0 to 0.0	0.0 to 113.3	-0.1 to -0.0
HDL	0.0 to 0.5	-0.0 to 0.0	0.0 to 0.4	-0.0 to 0.0	0.0 to 120.6	-0.7 to 0.8
HGB	0.0 to 7.2	-0.0 to 0.0	0.1 to 5.9	-0.0 to 0.0	0.1 to 113.1	-0.4 to -0.0
K	0.1 to 19.3	-0.0 to 0.0	0.2 to 16.9	-0.0 to 0.0	10.5 to 113.7	0.0 to 0.7
LDL	0.0 to 0.2	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 116.4	-1.5 to 0.9
MCH	0.0 to 7.2	-0.0 to 0.0	0.0 to 6.4	-0.0 to 0.0	0.0 to 115.9	-0.0 to 0.2
MCHC	0.0 to 9.7	-0.0 to 0.0	0.1 to 7.8	-0.0 to 0.0	0.8 to 113.6	-0.3 to -0.0
MCV	0.0 to 1.6	-0.0 to 0.0	0.0 to 0.8	-0.0 to 0.0	0.0 to 115.7	0.0 to 0.8
MPV	0.0 to 5.8	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 44.6	-0.3 to 0.5
NA	0.0 to 3.2	-0.0 to 0.0	0.0 to 1.8	-0.0 to 0.0	0.2 to 110.7	-0.0 to 0.1
PLT	0.0 to 0.1	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 111.0	-4.0 to -0.0
RBC	0.1 to 20.7	-0.0 to 0.0	0.2 to 17.2	-0.0 to 0.0	0.2 to 115.6	-0.3 to -0.0
RDW	0.0 to 5.3	-0.0 to 0.0	0.0 to 0.5	-0.0 to -0.0	0.0 to 137.5	-0.0 to 0.8
TBIL	0.1 to 36.2	-0.0 to 0.2	0.2 to 17.2	-0.1 to 0.1	0.2 to 121.3	-4.3 to -0.0
TC	0.0 to 0.3	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 115.0	-0.2 to 0.4
TGL	0.0 to 0.3	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 114.5	-0.0 to 1.8
TP	0.1 to 14.5	-0.0 to 0.0	0.1 to 12.6	-0.0 to 0.0	0.1 to 112.9	-0.2 to 0.0
WBC	0.0 to 4.5	-0.0 to 0.0	0.0 to 1.7	-0.0 to 0.0	0.0 to 109.7	-1.4 to -0.0
History Length (median [IQR])	4.8 [1.0, 14.4]	0.0 [0.0, 0.0]				
Prediction Horizon (median [IQR])			1.6 [0.1, 11.1]	0.0 [0.0, 0.0]		
Within-Person Variability (median [IQR])					115.2 [111.4, 120.5]	0.7 [0.3, 1.7

Table 4: Sensitivity Norma Quantile

	History Length		Prediction Horizon		Within-Person Variability	
Analyte	CI (% ref)	μ (%)	CI (% ref)	μ (%)	CI (% ref)	μ (
A1C	53.2 to 80.6	-25.0 to 3.8	63.4 to 74.4	-1.5 to 0.6	63.4 to 66.8	0.5 to
ALB	50.3 to 60.9	-11.5 to -0.0	52.0 to 59.8	-2.8 to -0.5	52.0 to 61.7	-2.2 to -
ALP	36.7 to 63.4	1.8 to 7.3	57.4 to 66.8	4.4 to 8.3	57.4 to 79.0	2.0 to
ALT	42.0 to 94.4	-29.6 to 2.9	71.9 to 85.5	-10.4 to 0.1	71.9 to 91.6	-3.6 to
AST	34.8 to 82.9	-27.8 to 3.7	63.8 to 81.3	-7.9 to 1.1	63.8 to 85.5	-2.5 to
BUN	42.9 to 90.8	-21.2 to 2.2	79.6 to 88.1	-1.8 to 1.5	80.3 to 88.3	-0.5 to
CA	78.2 to 106.2	-1.6 to 1.9	77.8 to 87.8	-0.1 to 0.5	79.3 to 84.2	0.1 to
CL	103.9 to 133.5	-2.1 to 0.7	103.7 to 104.1	0.2 to 0.3	104.1 to 104.5	0.3 to
CO2	89.4 to 116.9	-16.9 to 0.6	105.0 to 112.8	-2.2 to -0.7	106.4 to 108.8	-0.8 to -
CRE	81.6 to 87.5	-6.8 to 8.5	80.0 to 83.0	-10.6 to 2.2	82.5 to 86.4	-6.8 to -
DBIL	108.9 to 137.5	-44.1 to -6.6	126.4 to 135.2	-44.1 to -20.2	126.1 to 128.4	-44.4 to -3
GLU	42.7 to 103.7	1.0 to 7.7	97.0 to 99.9	1.3 to 2.1	97.5 to 105.6	0.0 to
HCT	87.3 to 152.6	-6.8 to 2.6	95.3 to 98.2	1.9 to 2.4	98.2 to 99.7	2.1 to
HDL	31.5 to 73.3	-2.1 to 14.9	68.9 to 75.7	-0.3 to 0.5	68.9 to 84.5	-1.7 to
HGB	58.7 to 79.1	-19.7 to -1.1	63.1 to 77.6	-3.2 to -1.8	63.1 to 69.1	-2.1 to -
K	80.3 to 89.5	-1.5 to 2.5	80.0 to 84.4	-0.9 to 1.7	80.3 to 86.6	-0.8 to -
LDL	33.9 to 55.1	-5.7 to 3.5	45.9 to 58.7	0.3 to 1.7	45.9 to 69.3	-0.9 to
MCH	66.8 to 105.4	-20.1 to 1.1	69.8 to 80.4	-0.4 to -0.0	71.4 to 78.2	-0.2 to
MCHC	72.9 to 114.8	-11.8 to 0.3	79.6 to 82.4	-0.3 to 0.1	80.7 to 84.4	0.0 to
MCV	57.3 to 69.9	-1.3 to 2.0	65.3 to 68.4	0.3 to 1.4	65.4 to 80.2	-0.1 to
MPV	12.2 to 31.8	-2.2 to 11.2	13.5 to 14.9	5.2 to 7.5	14.1 to 15.0	5.2 to
NA	94.9 to 101.7	0.1 to 0.7	94.9 to 95.9	0.1 to 0.2	94.9 to 95.6	0.1 to
PLT	85.4 to 109.4	-24.4 to -20.4	100.5 to 104.5	-20.5 to -20.5	102.2 to 122.2	-20.7 to -2
RBC	62.0 to 92.9	-5.9 to -1.1	62.0 to 69.7	-3.1 to -1.6	62.0 to 68.5	-2.7 to -
RDW	31.9 to 43.2	-8.2 to 3.7	31.7 to 37.6	1.1 to 2.8	32.4 to 41.2	2.8 to
TBIL	36.1 to 103.9	-50.9 to -8.9	76.7 to 99.4	-33.1 to -18.2	78.4 to 84.8	-32.2 to -2
TC	51.5 to 80.2	-5.6 to 0.2	72.4 to 75.4	-1.4 to 0.3	73.4 to 80.9	-2.4 to -
TGL	1.8 to 85.6	-35.2 to 17.4	75.4 to 80.1	4.3 to 6.3	75.5 to 88.2	-2.7 to
TP	67.8 to 101.2	-3.6 to 11.9	68.9 to 75.0	-0.1 to 1.1	70.7 to 82.1	0.2 to
WBC	56.2 to 82.9	-9.9 to 3.5	68.2 to 76.2	-1.0 to 3.9	69.0 to 79.5	-2.4 to
History Length (median [IQR])	28.3 [20.6, 41.8]	12.8 [5.6, 20.6]				
Prediction Horizon (median [IQR])			7.2 [3.0, 9.9]	1.6 [0.8, 3.2]		
Within-Person Variability (median [IQR])					7.2 [3.7, 12.4]	1.3 [0.3, 3

Table 5: Sensitivity Norma Gaussian

3.3 Validation — Combined

	eICU				CHS			
Analyte	N	CV_{intra}	CV_{inter}	II	N	CV_{intra}	CV_{inter}	II
A1C	—	—	—	—	336,494	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]
ALB	19,075	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	790,490	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]
ALP	13,582	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]	0.6 [0.5, 0.6]	1,099,044	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	0.4 [0.3, 0.4]
ALT	12,425	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]	0.6 [0.6, 0.6]	1,233,902	0.2 [0.2, 0.3]	0.5 [0.5, 0.5]	0.5 [0.5, 0.5]
AST	12,704	0.4 [0.4, 0.4]	0.5 [0.5, 0.5]	0.7 [0.7, 0.7]	1,236,781	0.2 [0.2, 0.2]	0.2 [0.2, 0.2]	0.8 [0.8, 0.8]
BUN	57,540	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]	0.6 [0.6, 0.6]	1,251,453	0.2 [0.2, 0.2]	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]
CA	57,067	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.8 [0.8, 0.8]	909,036	0.0 [0.0, 0.0]	0.1 [0.0, 0.1]	0.8 [0.8, 0.8]
CL	59,172	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.7, 0.7]	31,171	0.0 [0.0, 0.0]	0.1 [0.0, 0.1]	0.6 [0.6, 0.6]
CO2	70,208	0.1 [0.1, 0.1]	0.1 [0.1, 0.2]	0.8 [0.8, 0.8]	299	0.1 [0.1, 0.1]	0.3 [0.3, 0.4]	0.3 [0.3, 0.3]
CRE	54,820	0.2 [0.2, 0.2]	0.5 [0.5, 0.5]	0.4 [0.4, 0.4]	1,310,465	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]
DBIL	1,073	0.5 [0.4, 0.5]	0.7 [0.7, 0.7]	0.6 [0.6, 0.7]	815,774	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.8 [0.7, 0.8]
GLU	90,958	0.2 [0.2, 0.2]	0.2 [0.2, 0.2]	1.0 [1.0, 1.0]	1,351,168	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.3 [0.3, 0.4]
HCT	60,110	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,407,912	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.5, 0.5]
HDL	11	0.2 [0.2, 0.3]	0.5 [0.3, 0.6]	0.5 [0.3, 1.1]	1,002,204	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	0.4 [0.4, 0.4]
HGB	60,482	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,408,118	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.4, 0.5]
K	66,569	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	1.1 [1.1, 1.1]	1,182,544	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.9 [0.9, 0.9]
LDL	—	—	—	—	1,187,844	0.1 [0.1, 0.2]	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]
MCH	48,432	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,250,759	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]
MCHC	51,710	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.6 [0.6, 0.6]	1,407,568	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.9 [0.8, 0.9]
MCV	50,841	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,400,648	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.2 [0.2, 0.3]
MPV	37,083	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,407,716	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.6 [0.6, 0.6]
NA	124,270	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.8 [0.8, 0.8]	1,187,537	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.0 [1.0, 1.0]
PLT	53,371	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]	0.6 [0.6, 0.6]	1,407,716	0.1 [0.1, 0.2]	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]
RBC	53,584	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,404,794	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.4, 0.5]
RDW	48,715	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,403,339	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.7 [0.7, 0.7]
TBIL	12,738	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]	0.6 [0.6, 0.6]	815,774	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.8 [0.7, 0.8]
TC	35	0.2 [0.1, 0.2]	0.3 [0.2, 0.4]	0.5 [0.4, 0.8]	1,260,730	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.9 [0.9, 0.9]
TGL	312	0.3 [0.3, 0.3]	0.4 [0.4, 0.4]	0.7 [0.7, 0.8]	1,244,518	0.2 [0.2, 0.2]	0.2 [0.2, 0.3]	0.8 [0.8, 0.8]
TP	15,134	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.8 [0.8, 0.8]	791,481	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.7 [0.7, 0.7]
WBC	52,141	0.3 [0.3, 0.3]	0.3 [0.3, 0.3]	0.8 [0.8, 0.8]	1,409,027	0.2 [0.2, 0.2]	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]

Table 6: Variability

Dataset	N	Pop_{RI} (%)	Per_{RI} (%)	NORMA_{RI} (%)	Per_{RI} RR (%)	NORMA_{RI} RR (%)
eICU	29	50.2	34.9	45.7	37.3	9.1
CHS	30	6.8	29.3	18.0	21.3	6.6

Table 7: Prevalence

		Precision			Sensitivity			Specificity			
Dataset	Outcome	Per_{RI}	NORMA_{RI}	Δ	Per_{RI}	NORMA_{RI}	Δ	Per_{RI}	NORMA_{RI}	Δ	Per_{R}
eICU	Acute Kidney Injury	0.15	0.16	+0.01	0.41	0.21	-0.19	0.58	0.80	+0.21	0.5
	Mortality	0.10	0.13	+0.03	0.44	0.26	-0.18	0.59	0.80	+0.21	0.5
	Prolonged ICU Stay (>7d)	0.23	0.27	+0.04	0.38	0.25	-0.12	0.58	0.80	+0.22	0.5
	Sepsis	0.18	0.17	-0.00	0.41	0.20	-0.22	0.59	0.79	+0.21	0.5
CHS	Chronic Kidney Disease	0.09	0.13	+0.04	0.28	0.72	+0.45	0.57	0.83	+0.25	0.8
	Mortality	0.07	0.17	+0.10	0.27	0.73	+0.46	0.57	0.83	+0.26	0.6
	Type 2 Diabetes	0.08	0.15	+0.07	0.28	0.72	+0.44	0.57	0.83	+0.26	0.7

Table 8: Eval Summary

3.4 Validation — eICU

				Per$_{RI}$						
Outcome	Analyte	N	Events	Flagged	Precision	Sensitivity	Specificity	Accuracy	Flagged	Precision
Acute Kidney Injury	A1C	—	—	—	—	—	—	—	—	—
	ALB	891	130	218	0.14	0.23	0.75	0.68	110	0.17
	ALP	8,843	1,712	5,179	0.19	0.58	0.41	0.44	1,513	0.21
	ALT	7,391	1,408	4,191	0.17	0.51	0.42	0.44	1,467	0.18
	AST	8,035	1,384	4,187	0.17	0.50	0.47	0.48	1,203	0.16
	BUN	19,881	1,164	9,174	0.05	0.38	0.53	0.52	2,572	0.05
	CA	16,083	2,079	5,271	0.13	0.34	0.67	0.63	2,724	0.15
	CL	26,608	3,839	10,021	0.13	0.34	0.62	0.58	3,392	0.16
	CO2	29,891	3,601	9,186	0.09	0.23	0.68	0.63	4,976	0.11
	CRE	27,847	2,062	7,148	0.07	0.23	0.74	0.70	2,892	0.05
	DBIL	289	78	50	0.36	0.23	0.85	0.68	24	0.33
	GLU	9,534	1,257	2,366	0.11	0.20	0.74	0.67	832	0.16
	HCT	2,919	266	1,243	0.10	0.47	0.58	0.57	337	0.10
	HDL	—	—	—	—	—	—	—	—	—
	HGB	3,383	255	1,239	0.09	0.44	0.64	0.62	451	0.08
	K	57,801	8,390	19,896	0.14	0.34	0.65	0.61	9,968	0.17
	LDL	—	—	—	—	—	—	—	—	—
	MCH	32,416	4,932	15,125	0.16	0.48	0.54	0.53	8,045	0.17
	MCHC	22,846	3,339	8,833	0.14	0.37	0.61	0.58	3,557	0.14
	MCV	43,792	6,744	24,972	0.15	0.56	0.43	0.45	9,523	0.16
	MPV	34,479	5,045	18,234	0.15	0.53	0.47	0.48	30,053	0.15
	NA	40,769	6,154	17,705	0.14	0.40	0.56	0.53	25,674	0.15
	PLT	33,546	4,508	23,167	0.13	0.65	0.30	0.35	13,624	0.13
	RBC	4,595	429	1,255	0.11	0.31	0.73	0.69	545	0.11
	RDW	16,275	1,681	7,008	0.10	0.43	0.57	0.55	2,642	0.11
	TBIL	8,985	1,660	2,727	0.19	0.31	0.70	0.63	1,123	0.20
	TC	11	3	7	0.29	0.67	0.38	0.45	3	0.33
	TGL	176	42	96	0.21	0.48	0.43	0.44	27	0.22
	TP	4,859	864	1,642	0.19	0.36	0.67	0.61	680	0.21
	WBC	25,827	3,491	10,890	0.12	0.39	0.57	0.55	3,334	0.14
Mortality	A1C	—	—	—	—	—	—	—	—	—
	ALB	891	49	218	0.05	0.20	0.75	0.72	110	0.08
	ALP	8,843	1,135	5,179	0.13	0.60	0.42	0.44	1,513	0.16
	ALT	7,391	909	4,191	0.12	0.56	0.43	0.45	1,467	0.12
	AST	8,035	786	4,187	0.10	0.53	0.48	0.48	1,203	0.11
	BUN	19,881	1,043	9,174	0.05	0.48	0.54	0.54	2,572	0.06
	CA	16,083	1,199	5,271	0.08	0.34	0.67	0.65	2,724	0.08
	CL	26,608	2,122	10,021	0.08	0.36	0.62	0.60	3,392	0.10
	CO2	29,891	2,365	9,186	0.06	0.23	0.69	0.65	4,976	0.08
	CRE	27,847	1,863	7,148	0.08	0.31	0.75	0.72	2,892	0.10
	DBIL	289	49	50	0.06	0.06	0.80	0.68	24	0.04
	GLU	9,534	688	2,366	0.04	0.15	0.74	0.70	832	0.08
	HCT	2,919	250	1,243	0.10	0.49	0.58	0.57	337	0.13
	HDL	—	—	—	—	—	—	—	—	—
	HGB	3,383	260	1,239	0.09	0.42	0.64	0.62	451	0.12
	K	57,801	5,617	19,896	0.11	0.38	0.66	0.63	9,968	0.12
	LDL	—	—	—	—	—	—	—	—	—
	MCH	32,416	3,229	15,125	0.11	0.49	0.54	0.53	8,045	0.12
	MCHC	22,846	1,971	8,833	0.09	0.41	0.62	0.60	3,557	0.10
	MCV	43,792	4,275	24,972	0.10	0.61	0.43	0.45	9,523	0.13
	MPV	34,479	3,508	18,234	0.11	0.59	0.48	0.49	30,053	0.11
	NA	40,769	3,781	17,705	0.09	0.44	0.57	0.55	25,674	0.10
	PLT	33,546	2,550	23,167	0.07	0.65	0.31	0.33	13,624	0.07
	RBC	4,595	350	1,255	0.10	0.35	0.73	0.70	545	0.10
	RDW	16,275	874	7,008	0.06	0.49	0.57	0.57	2,642	0.06
	TBIL	8,985	1,035	2,727	0.13	0.35	0.70	0.66	1,123	0.15
	TC	11	2	7	0.29	1.00	0.44	0.55	3	0.67
	TGL	176	34	96	0.20	0.56	0.46	0.48	27	0.22
	TP	4,859	484	1,642	0.11	0.38	0.67	0.64	680	0.11
	WBC	25,827	1,695	10,890	0.07	0.42	0.58	0.57	3,334	0.07
Prolonged ICU Stay (>7d)	A1C	—	—	—	—	—	—	—	—	—
	ALB	891	158	218	0.17	0.24	0.75	0.66	110	0.23
	ALP	8,843	2,882	5,179	0.32	0.57	0.41	0.46	1,513	0.35
	ALT	7,391	2,412	4,191	0.31	0.55	0.42	0.46	1,467	0.33
	AST	8,035	2,561	4,187	0.30	0.49	0.46	0.47	1,203	0.31

Table 10: Cox PH: Mortality

			Pop_{RI}			Per_{RI}			$NORMA_{RI}$		
Analyte	N train (events)	N test (events)	HR [95% CI]	C		HR [95% CI]	C		HR [95% CI]	C	
A1C	—	—	—	—		—	—		—	—	
ALB	11152 (1546)	7435 (1030)	1.52 [1.03, 2.22]	0.519	*	1.34 [1.21, 1.48]	0.541	*	1.81 [1.48, 2.20]	0.529	*
ALP	7966 (1112)	5312 (741)	0.94 [0.83, 1.06]	0.532		1.18 [1.04, 1.35]	0.532	*	1.12 [1.00, 1.27]	0.536	
ALT	7288 (968)	4859 (646)	1.01 [0.89, 1.15]	0.534		1.11 [0.97, 1.27]	0.542		1.09 [0.96, 1.24]	0.533	
AST	7446 (904)	4964 (603)	1.81 [1.58, 2.07]	0.582	*	1.33 [1.16, 1.53]	0.567	*	1.65 [1.44, 1.88]	0.583	*
BUN	33377 (3337)	22252 (2224)	1.85 [1.67, 2.05]	0.578	*	1.55 [1.44, 1.67]	0.581	*	1.95 [1.78, 2.13]	0.588	*
CA	33079 (3432)	22053 (2288)	1.70 [1.55, 1.87]	0.579	*	1.16 [1.08, 1.24]	0.548	*	1.69 [1.55, 1.85]	0.580	*
CL	34318 (3515)	22880 (2344)	1.42 [1.31, 1.53]	0.591	*	1.26 [1.18, 1.35]	0.581	*	1.54 [1.43, 1.66]	0.588	*
CO2	40678 (4737)	27119 (3158)	2.02 [1.84, 2.23]	0.587	*	1.05 [0.99, 1.12]	0.557		1.82 [1.67, 1.98]	0.580	*
CRE	31721 (3112)	21148 (2074)	1.77 [1.63, 1.92]	0.606	*	1.78 [1.66, 1.91]	0.600	*	1.83 [1.70, 1.98]	0.612	*
DBIL	684 (121)	456 (81)	1.41 [0.91, 2.19]	0.514		1.14 [0.79, 1.65]	0.492		1.78 [1.23, 2.58]	0.586	*
GLU	52612 (5123)	35076 (3416)	2.18 [1.77, 2.70]	0.565	*	1.05 [0.99, 1.11]	0.559		1.47 [1.28, 1.69]	0.558	*
HCT	34808 (3508)	23206 (2338)	1.16 [0.92, 1.45]	0.557		1.34 [1.25, 1.43]	0.580	*	1.32 [1.12, 1.54]	0.557	*
HDL	—	—	—	—		—	—		—	—	
HGB	35017 (3578)	23346 (2385)	1.13 [0.92, 1.40]	0.549		1.40 [1.31, 1.50]	0.571	*	1.29 [1.10, 1.50]	0.550	*
K	38601 (3892)	25734 (2594)	1.35 [1.26, 1.44]	0.565	*	1.31 [1.23, 1.39]	0.566	*	1.31 [1.23, 1.40]	0.556	*
LDL	—	—	—	—		—	—		—	—	
MCH	28066 (2850)	18712 (1900)	1.01 [0.93, 1.08]	0.555		1.02 [0.95, 1.10]	0.555		1.04 [0.97, 1.13]	0.554	
MCHC	29986 (3025)	19992 (2016)	0.85 [0.78, 0.93]	0.561	*	1.27 [1.18, 1.36]	0.572	*	0.89 [0.82, 0.97]	0.559	*
MCV	29487 (2977)	19659 (1985)	1.21 [1.11, 1.31]	0.565	*	1.33 [1.24, 1.44]	0.567	*	1.34 [1.24, 1.44]	0.573	*
MPV	21531 (2154)	14355 (1436)	0.94 [0.83, 1.06]	0.565		1.26 [1.15, 1.37]	0.574	*	1.21 [1.04, 1.41]	0.567	*
NA	36225 (3788)	24151 (2525)	1.20 [1.12, 1.29]	0.554	*	1.33 [1.25, 1.43]	0.567	*	1.21 [1.11, 1.33]	0.552	*
PLT	30955 (3170)	20637 (2113)	2.46 [2.29, 2.65]	0.657	*	0.92 [0.85, 0.99]	0.550	*	1.64 [1.52, 1.78]	0.579	*
RBC	31070 (3180)	20714 (2120)	1.05 [0.88, 1.25]	0.555		1.35 [1.26, 1.44]	0.572	*	1.20 [1.05, 1.38]	0.559	*
RDW	28255 (2815)	18837 (1877)	1.83 [1.63, 2.05]	0.591	*	1.29 [1.20, 1.40]	0.574	*	1.57 [1.42, 1.74]	0.582	*
TBIL	7466 (925)	4978 (617)	1.43 [1.25, 1.63]	0.561	*	1.36 [1.19, 1.54]	0.569	*	1.61 [1.41, 1.83]	0.567	*
TC	—	—	—	—		—	—		—	—	
TGL	186 (37)	124 (24)	1.09 [0.54, 2.18]	0.594		0.93 [0.47, 1.84]	0.595		1.39 [0.70, 2.79]	0.588	
TP	8881 (1298)	5922 (865)	1.78 [1.54, 2.07]	0.566	*	1.02 [0.92, 1.14]	0.533		1.84 [1.61, 2.09]	0.582	*
WBC	30249 (2881)	20167 (1921)	2.15 [1.95, 2.36]	0.619	*	1.56 [1.44, 1.68]	0.586	*	2.01 [1.84, 2.19]	0.609	*

Table 11: Cox PH: Acute Kidney Injury

			Pop_{RI}			Per_{RI}			$NORMA_{RI}$		
Analyte	N train (events)	N test (events)	HR [95% CI]	C		HR [95% CI]	C		HR [95% CI]	C	
A1C	—	—	—	—		—	—		—	—	
ALB	11121 (2490)	7414 (1660)	1.28 [1.01, 1.63]	0.516	*	1.12 [1.04, 1.21]	0.508	*	1.31 [1.14, 1.51]	0.521	*
ALP	7949 (1702)	5300 (1134)	1.25 [1.14, 1.37]	0.544	*	0.97 [0.88, 1.07]	0.505		1.22 [1.11, 1.34]	0.527	*
ALT	7267 (1472)	4846 (982)	1.08 [0.97, 1.19]	0.504		0.89 [0.80, 0.99]	0.515	*	0.98 [0.89, 1.09]	0.505	
AST	7429 (1464)	4954 (976)	1.19 [1.08, 1.32]	0.529	*	0.97 [0.87, 1.08]	0.505		1.11 [1.00, 1.23]	0.506	*
BUN	33277 (4964)	22186 (3309)	1.87 [1.73, 2.01]	0.567	*	0.95 [0.90, 1.01]	0.518		1.19 [1.12, 1.27]	0.531	*
CA	32978 (5406)	21986 (3604)	1.42 [1.33, 1.52]	0.542	*	1.04 [0.99, 1.10]	0.528		1.29 [1.22, 1.37]	0.539	*
CL	34211 (5470)	22808 (3647)	1.43 [1.35, 1.52]	0.542	*	0.98 [0.93, 1.04]	0.522		1.32 [1.25, 1.40]	0.542	*
CO2	40564 (6163)	27043 (4109)	1.38 [1.29, 1.47]	0.538	*	0.97 [0.92, 1.02]	0.526		1.16 [1.09, 1.23]	0.525	*
CRE	31636 (4453)	21091 (2969)	1.78 [1.66, 1.89]	0.571	*	1.18 [1.11, 1.25]	0.531	*	1.15 [1.09, 1.22]	0.522	*
DBIL	682 (211)	456 (141)	1.32 [0.97, 1.79]	0.520		1.18 [0.90, 1.56]	0.502		1.32 [1.01, 1.74]	0.503	*
GLU	52473 (6865)	34982 (4577)	1.09 [0.96, 1.23]	0.522		1.00 [0.95, 1.05]	0.522		1.08 [0.99, 1.19]	0.523	
HCT	34714 (5014)	23144 (3343)	1.86 [1.51, 2.29]	0.522	*	1.02 [0.97, 1.08]	0.517		1.36 [1.20, 1.54]	0.522	*
HDL	—	—	—	—		—	—		—	—	
HGB	34921 (5008)	23282 (3339)	2.11 [1.72, 2.59]	0.526	*	1.03 [0.97, 1.09]	0.522		1.50 [1.32, 1.70]	0.524	*
K	38493 (5646)	25662 (3764)	1.43 [1.35, 1.50]	0.554	*	1.11 [1.06, 1.17]	0.531	*	1.34 [1.27, 1.41]	0.547	*
LDL	—	—	—	—		—	—		—	—	
MCH	27979 (4312)	18654 (2875)	1.12 [1.05, 1.19]	0.528	*	1.06 [0.99, 1.12]	0.526		1.11 [1.05, 1.18]	0.530	*
MCHC	29897 (4620)	19932 (3080)	1.18 [1.11, 1.26]	0.530	*	1.02 [0.97, 1.08]	0.525		1.08 [1.02, 1.15]	0.529	*
MCV	29399 (4527)	19600 (3018)	1.14 [1.05, 1.22]	0.523	*	1.00 [0.94, 1.06]	0.523		1.08 [1.01, 1.14]	0.523	*
MPV	21458 (3143)	14306 (2096)	1.28 [1.17, 1.41]	0.534	*	0.98 [0.91, 1.05]	0.518		1.08 [0.96, 1.21]	0.523	
NA	36121 (5599)	24081 (3732)	1.11 [1.05, 1.17]	0.528	*	0.97 [0.92, 1.02]	0.524		0.95 [0.89, 1.02]	0.525	
PLT	30865 (4734)	20577 (3156)	1.39 [1.31, 1.47]	0.545	*	0.87 [0.82, 0.93]	0.525	*	1.13 [1.06, 1.20]	0.527	*
RBC	30979 (4736)	20654 (3157)	1.76 [1.51, 2.06]	0.521	*	1.04 [0.98, 1.10]	0.521		1.45 [1.30, 1.62]	0.528	*
RDW	28173 (4340)	18782 (2893)	1.74 [1.61, 1.88]	0.571	*	1.15 [1.08, 1.22]	0.537	*	1.66 [1.54, 1.79]	0.565	*
TBIL	7447 (1482)	4966 (988)	1.15 [1.04, 1.28]	0.526	*	1.08 [0.97, 1.19]	0.518		1.07 [0.97, 1.19]	0.534	
TC	20 (4)	14 (2)	0.52 [0.07, 3.77]	0.640		1.43 [0.13, 16.07]	0.600		0.41 [0.04, 3.84]	0.600	
TGL	186 (54)	124 (36)	1.18 [0.67, 2.09]	0.551		1.20 [0.67, 2.16]	0.530		0.91 [0.52, 1.56]	0.489	
TP	8860 (1890)	5908 (1260)	1.28 [1.15, 1.43]	0.517	*	1.09 [1.00, 1.20]	0.499		1.22 [1.10, 1.34]	0.527	*
WBC	30160 (4531)	20108 (3021)	1.26 [1.18, 1.34]	0.540	*	0.99 [0.94, 1.05]	0.525		1.21 [1.14, 1.28]	0.536	*

Table 12: Cox PH: Sepsis

			Pop_{RI}			Per_{RI}			NORMA_{RI}		
Analyte	N train (events)	N test (events)	HR [95% CI]	C		HR [95% CI]	C		HR [95% CI]	C	
A1C	—	—	—	—		—	—		—	—	
ALB	11106 (2754)	7405 (1836)	2.27 [1.72, 3.00]	0.519	*	1.14 [1.05, 1.22]	0.518	*	1.25 [1.10, 1.42]	0.519	*
ALP	7938 (2020)	5292 (1347)	1.17 [1.07, 1.28]	0.544	*	0.87 [0.80, 0.95]	0.504	*	1.12 [1.02, 1.22]	0.524	*
ALT	7256 (1877)	4838 (1252)	1.01 [0.92, 1.11]	0.510		0.99 [0.90, 1.10]	0.514		0.97 [0.88, 1.06]	0.514	
AST	7415 (1904)	4944 (1270)	1.08 [0.98, 1.18]	0.502		0.96 [0.88, 1.06]	0.522		1.07 [0.98, 1.17]	0.489	
BUN	33238 (5689)	22160 (3793)	1.22 [1.15, 1.30]	0.534	*	0.90 [0.85, 0.95]	0.525	*	1.01 [0.95, 1.06]	0.527	
CA	32940 (5801)	21960 (3868)	1.54 [1.45, 1.65]	0.546	*	1.09 [1.03, 1.14]	0.520	*	1.40 [1.32, 1.49]	0.536	*
CL	34174 (5908)	22784 (3939)	1.33 [1.26, 1.41]	0.540	*	0.95 [0.90, 1.00]	0.525		1.21 [1.15, 1.28]	0.533	*
CO2	40519 (6797)	27014 (4532)	1.28 [1.20, 1.36]	0.543	*	0.94 [0.89, 0.99]	0.527	*	1.11 [1.05, 1.17]	0.533	*
CRE	31593 (5313)	21063 (3542)	1.28 [1.21, 1.35]	0.544	*	1.08 [1.02, 1.14]	0.523	*	1.03 [0.98, 1.09]	0.522	
DBIL	682 (186)	456 (125)	0.98 [0.72, 1.33]	0.513		1.33 [1.00, 1.78]	0.505		1.34 [1.00, 1.79]	0.499	*
GLU	52426 (7601)	34951 (5068)	0.98 [0.88, 1.09]	0.528		0.95 [0.91, 0.99]	0.529	*	0.91 [0.84, 0.99]	0.527	*
HCT	34669 (5743)	23114 (3829)	1.54 [1.29, 1.84]	0.527	*	0.99 [0.94, 1.05]	0.526		1.22 [1.10, 1.36]	0.527	*
HDL	—	—	—	—		—	—		—	—	
HGB	34875 (5786)	23251 (3858)	1.54 [1.31, 1.80]	0.528	*	0.98 [0.93, 1.03]	0.522		1.26 [1.13, 1.40]	0.524	*
K	38458 (6287)	25640 (4192)	1.46 [1.39, 1.54]	0.562	*	1.04 [0.99, 1.09]	0.531		1.37 [1.31, 1.44]	0.547	*
LDL	—	—	—	—		—	—		—	—	
MCH	27942 (5009)	18629 (3340)	1.21 [1.14, 1.27]	0.533	*	1.06 [1.00, 1.12]	0.519	*	1.19 [1.13, 1.26]	0.531	*
MCHC	29859 (5362)	19907 (3575)	1.36 [1.27, 1.44]	0.540	*	1.01 [0.96, 1.07]	0.517		1.26 [1.18, 1.34]	0.535	*
MCV	29361 (5219)	19574 (3480)	1.24 [1.16, 1.33]	0.530	*	1.04 [0.99, 1.10]	0.511		1.20 [1.14, 1.27]	0.520	*
MPV	21431 (3422)	14288 (2282)	1.25 [1.14, 1.37]	0.520	*	0.99 [0.93, 1.06]	0.516		1.04 [0.93, 1.16]	0.514	
NA	36084 (6029)	24056 (4020)	1.05 [1.00, 1.11]	0.519	*	0.92 [0.88, 0.97]	0.521	*	1.00 [0.94, 1.06]	0.519	
PLT	30825 (5403)	20551 (3602)	1.16 [1.10, 1.23]	0.532	*	0.92 [0.86, 0.98]	0.526	*	1.02 [0.97, 1.08]	0.528	
RBC	30937 (5514)	20626 (3677)	1.33 [1.17, 1.51]	0.525	*	1.01 [0.95, 1.06]	0.517		1.24 [1.12, 1.36]	0.522	*
RDW	28134 (5022)	18757 (3348)	1.80 [1.68, 1.93]	0.554	*	1.15 [1.09, 1.22]	0.515	*	1.61 [1.50, 1.72]	0.551	*
TBIL	7437 (1878)	4959 (1252)	1.02 [0.93, 1.12]	0.517		1.00 [0.91, 1.10]	0.513		1.02 [0.93, 1.12]	0.514	
TC	20 (6)	14 (5)	0.71 [0.13, 3.89]	0.425		1.73 [0.27, 11.17]	0.550		0.31 [0.04, 2.18]	0.600	
TGL	186 (63)	124 (42)	0.95 [0.56, 1.59]	0.409		1.16 [0.67, 2.00]	0.431		0.76 [0.46, 1.26]	0.403	
TP	8848 (2278)	5899 (1518)	1.07 [0.97, 1.18]	0.529		1.13 [1.04, 1.23]	0.519	*	1.04 [0.95, 1.14]	0.528	
WBC	30126 (5169)	20085 (3447)	1.49 [1.40, 1.58]	0.544	*	0.93 [0.88, 0.98]	0.523	*	1.28 [1.21, 1.35]	0.530	*

Table 13: Cox PH: Prolonged ICU Stay (>7d)

			Pop_{RI}			Per_{RI}			$NORMA_{RI}$		
Analyte	N train (events)	N test (events)	HR [95% CI]	C		HR [95% CI]	C		HR [95% CI]	C	
A1C	—	—	—	—		—	—		—	—	
ALB	11140 (3587)	7428 (2392)	0.75 [0.61, 0.93]	0.535	*	0.88 [0.82, 0.94]	0.537	*	1.17 [1.05, 1.30]	0.539	*
ALP	7956 (2821)	5305 (1881)	0.68 [0.63, 0.73]	0.551	*	0.97 [0.90, 1.05]	0.524		0.80 [0.75, 0.87]	0.544	*
ALT	7276 (2596)	4851 (1731)	0.82 [0.76, 0.88]	0.533	*	1.00 [0.92, 1.08]	0.519		0.88 [0.82, 0.96]	0.520	*
AST	7437 (2658)	4958 (1772)	0.86 [0.80, 0.93]	0.545	*	1.03 [0.95, 1.11]	0.530		0.88 [0.82, 0.95]	0.542	*
BUN	33334 (7040)	22224 (4693)	0.67 [0.63, 0.71]	0.551	*	1.01 [0.96, 1.06]	0.530		0.70 [0.66, 0.74]	0.551	*
CA	33037 (6995)	22025 (4664)	0.79 [0.75, 0.83]	0.543	*	0.90 [0.86, 0.95]	0.537	*	0.80 [0.76, 0.84]	0.546	*
CL	34276 (7201)	22852 (4801)	0.73 [0.69, 0.77]	0.549	*	0.99 [0.95, 1.04]	0.533		0.77 [0.73, 0.81]	0.546	*
CO2	40635 (7192)	27090 (4795)	0.64 [0.59, 0.69]	0.548	*	0.98 [0.93, 1.03]	0.533		0.65 [0.61, 0.70]	0.548	*
CRE	31682 (6787)	21122 (4524)	0.68 [0.65, 0.72]	0.566	*	0.80 [0.76, 0.84]	0.552	*	0.78 [0.74, 0.81]	0.561	*
DBIL	684 (328)	456 (218)	1.02 [0.81, 1.29]	0.512		1.07 [0.86, 1.34]	0.490		1.15 [0.93, 1.44]	0.527	
GLU	52558 (7372)	35039 (4915)	0.43 [0.36, 0.52]	0.536	*	0.84 [0.80, 0.88]	0.536	*	0.44 [0.38, 0.51]	0.540	*
HCT	34764 (7128)	23177 (4752)	0.50 [0.43, 0.58]	0.543	*	0.97 [0.92, 1.01]	0.538		0.64 [0.58, 0.71]	0.546	*
HDL	—	—	—	—		—	—		—	—	
HGB	34971 (7127)	23315 (4752)	0.49 [0.43, 0.57]	0.546	*	0.96 [0.92, 1.01]	0.539		0.62 [0.56, 0.69]	0.543	*
K	38556 (7280)	25705 (4853)	0.76 [0.73, 0.80]	0.551	*	0.88 [0.84, 0.92]	0.537	*	0.76 [0.72, 0.79]	0.555	*
LDL	—	—	—	—		—	—		—	—	
MCH	28024 (6450)	18684 (4301)	0.91 [0.87, 0.96]	0.535	*	0.83 [0.79, 0.87]	0.546	*	0.84 [0.80, 0.89]	0.539	*
MCHC	29943 (6866)	19963 (4578)	0.66 [0.62, 0.70]	0.559	*	0.92 [0.88, 0.96]	0.540	*	0.65 [0.61, 0.69]	0.561	*
MCV	29444 (6783)	19630 (4522)	0.89 [0.84, 0.94]	0.541	*	0.99 [0.94, 1.04]	0.539		0.85 [0.81, 0.89]	0.545	*
MPV	21490 (4994)	14328 (3330)	0.84 [0.78, 0.91]	0.533	*	0.98 [0.92, 1.03]	0.527		0.93 [0.85, 1.03]	0.530	
NA	36184 (7220)	24124 (4813)	0.74 [0.70, 0.78]	0.559	*	0.98 [0.93, 1.03]	0.539		0.76 [0.72, 0.81]	0.547	*
PLT	30912 (7029)	20608 (4686)	0.90 [0.86, 0.95]	0.535	*	1.23 [1.17, 1.30]	0.541	*	0.98 [0.93, 1.03]	0.536	
RBC	31026 (7104)	20685 (4737)	0.55 [0.48, 0.61]	0.549	*	0.87 [0.83, 0.91]	0.536	*	0.64 [0.58, 0.70]	0.549	*
RDW	28215 (6495)	18811 (4331)	0.67 [0.63, 0.71]	0.554	*	0.85 [0.81, 0.90]	0.537	*	0.70 [0.66, 0.75]	0.554	*
TBIL	7456 (2547)	4972 (1699)	0.86 [0.79, 0.93]	0.537	*	0.94 [0.87, 1.01]	0.536		0.95 [0.87, 1.02]	0.531	
TC	20 (8)	14 (5)	0.67 [0.11, 4.11]	0.600		3.21 [0.25, 40.70]	0.200		1.25 [0.21, 7.47]	0.300	
TGL	186 (134)	124 (90)	0.82 [0.57, 1.18]	0.527		0.93 [0.64, 1.35]	0.519		0.87 [0.61, 1.23]	0.531	
TP	8871 (3098)	5914 (2065)	1.03 [0.95, 1.12]	0.524		0.92 [0.86, 0.99]	0.529	*	1.07 [0.99, 1.15]	0.528	
WBC	30208 (7002)	20139 (4668)	0.73 [0.69, 0.77]	0.553	*	1.01 [0.96, 1.06]	0.532		0.76 [0.72, 0.80]	0.548	*

Analyte	NORMA-only flags	Later Pop_{RI} flag	Median lead (h)	IQR (h)
A1C	—	—	—	—
ALB	458	119	26.5	22.8–54.0
ALP	2,722	374	35.2	23.7–73.1
ALT	2,840	432	24.9	23.4–51.5
AST	2,070	319	24.9	23.3–50.6
BUN	5,767	1,209	24.6	22.1–48.2
CA	7,879	1,508	24.9	19.0–51.5
CL	7,724	2,248	24.8	22.3–48.6
CO2	15,069	6,065	24.5	18.7–48.6
CRE	5,285	766	24.4	20.3–48.0
DBIL	61	2	46.2	35.1–57.2
GLU	5,325	3,510	6.8	3.5–18.2
HCT	887	227	23.4	13.5–29.0
HDL	0	0	—	—
HGB	1,038	226	23.7	16.7–29.0
K	17,062	4,193	24.8	18.4–50.4
LDL	0	0	—	—
MCH	14,778	1,360	33.3	23.3–72.5
MCHC	6,131	1,122	25.7	22.9–54.8
MCV	18,737	917	46.3	23.6–76.9
MPV	85,001	3,941	29.6	22.8–71.7
NA	208,205	55,713	26.4	20.7–68.7
PLT	35,602	3,266	43.0	23.7–72.2
RBC	1,192	256	24.3	20.8–47.7
RDW	4,586	367	26.3	23.5–53.5
TBIL	2,104	227	27.4	23.2–69.6
TC	8	2	122.4	98.4–146.3
TGL	43	5	37.8	24.2–47.1
TP	1,918	334	29.6	23.4–88.7
WBC	6,250	1,285	24.7	22.5–49.4
Overall	458,742	89,993	25.7	22.8–51.5

Table 14: Lead Time

3.5 Validation — CHS

				Per_{RI}						
Outcome	Analyte	N	Events	Flagged	Precision	Sensitivity	Specificity	Accuracy	Flagged	Precision
Chronic Kidney Disease	A1C	200,000	10,000	32,110	0.11	0.35	0.62	0.82	82,278	0.08
	ALB	200,000	10,000	31,645	0.08	0.25	0.56	0.82	50,335	0.15
	ALP	200,000	10,000	28,571	0.07	0.20	0.51	0.83	47,058	0.17
	ALT	200,000	10,000	28,767	0.07	0.21	0.52	0.83	48,466	0.16
	AST	200,000	10,000	32,894	0.08	0.25	0.56	0.81	48,076	0.16
	BUN	200,000	10,000	29,072	0.06	0.16	0.49	0.82	40,975	0.20
	CA	200,000	10,000	28,301	0.11	0.30	0.62	0.84	81,395	0.09
	CL	200,000	10,000	31,553	0.10	0.33	0.61	0.82	72,580	0.09
	CO2	200,000	10,000	32,514	0.10	0.33	0.61	0.82	71,505	0.09
	CRE	200,000	10,000	32,608	0.05	0.15	0.50	0.80	38,053	0.23
	DBIL	200,000	10,000	33,027	0.11	0.36	0.63	0.82	81,012	0.08
	GLU	200,000	10,000	35,156	0.06	0.22	0.55	0.80	42,119	0.18
	HCT	200,000	10,000	28,658	0.08	0.23	0.53	0.83	53,873	0.14
	HDL	200,000	10,000	31,958	0.10	0.31	0.59	0.82	64,476	0.11
	HGB	200,000	10,000	27,439	0.08	0.22	0.54	0.84	54,577	0.14
	K	200,000	10,000	30,927	0.10	0.30	0.60	0.83	65,420	0.11
	LDL	200,000	10,000	32,000	0.10	0.32	0.59	0.82	67,990	0.10
	MCH	200,000	10,000	29,545	0.09	0.26	0.55	0.83	57,812	0.13
	MCHC	200,000	10,000	27,500	0.10	0.28	0.61	0.84	72,500	0.10
	MCV	200,000	10,000	28,723	0.09	0.27	0.57	0.83	64,035	0.11
	MPV	200,000	10,000	33,482	0.11	0.38	0.64	0.82	86,805	0.07
	NA	200,000	10,000	32,066	0.11	0.34	0.62	0.82	76,744	0.09
	PLT	200,000	10,000	28,571	0.09	0.26	0.57	0.83	61,157	0.12
	RBC	200,000	10,000	30,113	0.09	0.27	0.53	0.83	57,421	0.13
	RDW	200,000	10,000	28,235	0.09	0.24	0.56	0.83	56,296	0.14
	TBIL	200,000	10,000	31,818	0.09	0.28	0.58	0.82	56,250	0.13
	TC	200,000	10,000	32,978	0.09	0.31	0.59	0.82	60,517	0.11
	TGL	200,000	10,000	32,989	0.10	0.32	0.60	0.82	63,542	0.11
	TP	200,000	10,000	32,352	0.09	0.28	0.57	0.82	53,703	0.14
	WBC	200,000	10,000	31,868	0.09	0.29	0.58	0.82	58,677	0.12
Mortality	A1C	200,000	16,000	48,000	0.10	0.30	0.61	0.73	112,000	0.10
	ALB	200,000	16,000	81,538	0.05	0.27	0.53	0.55	55,471	0.21
	ALP	200,000	16,000	75,999	0.04	0.19	0.49	0.57	54,000	0.24
	ALT	200,000	16,000	94,117	0.03	0.20	0.50	0.48	50,393	0.25
	AST	200,000	16,000	86,956	0.05	0.25	0.54	0.53	53,097	0.23
	BUN	200,000	16,000	85,970	0.07	0.36	0.52	0.55	57,853	0.18
	CA	200,000	16,000	58,352	0.09	0.31	0.61	0.68	81,777	0.14
	CL	200,000	16,000	37,103	0.10	0.22	0.60	0.77	115,887	0.11
	CO2	200,000	16,000	65,934	0.09	0.38	0.60	0.65	82,644	0.12
	CRE	200,000	16,000	84,363	0.06	0.29	0.51	0.54	55,414	0.20
	DBIL	200,000	16,000	49,361	0.09	0.29	0.62	0.72	99,649	0.11
	GLU	200,000	16,000	69,508	0.06	0.27	0.52	0.61	61,570	0.19
	HCT	200,000	16,000	62,051	0.06	0.22	0.56	0.65	62,626	0.20
	HDL	200,000	16,000	54,736	0.08	0.26	0.58	0.69	75,897	0.16
	HGB	200,000	16,000	43,750	0.06	0.17	0.56	0.73	71,739	0.18
	K	200,000	16,000	63,636	0.09	0.35	0.59	0.66	81,250	0.13
	LDL	200,000	16,000	58,947	0.08	0.28	0.58	0.67	73,846	0.16
	MCH	200,000	16,000	65,573	0.06	0.25	0.57	0.63	62,827	0.19
	MCHC	200,000	16,000	30,379	0.08	0.15	0.60	0.79	91,275	0.15
	MCV	200,000	16,000	50,731	0.08	0.26	0.59	0.71	83,380	0.14
	MPV	200,000	16,000	51,376	0.11	0.35	0.62	0.72	131,645	0.08
	NA	200,000	16,000	49,056	0.11	0.33	0.61	0.73	125,581	0.09
	PLT	200,000	16,000	134,399	0.03	0.21	0.57	0.28	45,963	0.28
	RBC	200,000	16,000	76,557	0.07	0.33	0.56	0.59	62,588	0.17
	RDW	200,000	16,000	60,000	0.06	0.24	0.57	0.66	66,086	0.18
	TBIL	200,000	16,000	95,652	0.05	0.28	0.55	0.49	51,327	0.23
	TC	200,000	16,000	89,795	0.05	0.28	0.55	0.52	52,968	0.22
	TGL	200,000	16,000	62,857	0.07	0.28	0.58	0.65	68,235	0.17
	TP	200,000	16,000	62,857	0.07	0.28	0.56	0.65	68,235	0.17
	WBC	200,000	16,000	65,753	0.07	0.30	0.58	0.64	68,711	0.16
Type 2 Diabetes	A1C	200,000	12,000	56,441	0.03	0.16	0.47	0.68	39,685	0.25
	ALB	200,000	12,000	48,750	0.06	0.26	0.54	0.73	48,260	0.18
	ALP	200,000	12,000	45,818	0.06	0.21	0.50	0.74	46,243	0.20
	ALT	200,000	12,000	51,903	0.05	0.22	0.51	0.71	43,867	0.21
	AST	200,000	12,000	51,724	0.06	0.25	0.53	0.71	45,454	0.20